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# On unknotting tunnels for knots

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Let $K$ be a knot in a closed orientable 3-manifold $M$, and $E(K)$ be the exterior of $K$, that is, the complement of an open tubular neighbourhood of $K$. In this paper, we use the term *tunnel* to mean a properly embedded arc in $E(K)$. Two tunnels $\tau_1$ and $\tau_2$ are said to be *isotopic* (*homeomorphic* resp.), if there is an ambient isotopy (homeomorphism resp.) of $E(K)$ which sends $\tau_1$ to $\tau_2$. A tunnel $\tau$ for $K$ is called an *unknotting tunnel*, if $\mathrm{cl}(E(K)-N(\tau; E(K)))$ is a genus two handlebody. Here $N(X; Y)$ denotes a regular neighbourhood of $X$ in $Y$, and $\mathrm{cl}(X)$ denotes the closure of $X$.

The concept of an unknotting tunnel is closely related to a *genus 2 Heegaard decomposition* of $E(K)$; that is, a decomposition of $E(K)$ into a genus 2 handle-body and a compression-body (cf. [BRZ, BM]). In fact, each unknotting tunnel $\tau$ determines a genus 2 Heegaard decomposition $(V_1(\tau), V_2(\tau))$ of $E(K)$, where $V_2(\tau) = N(\partial E(K) \cup \tau; E(K))$ and $V_1(\tau) = \mathrm{cl}(E(K)-V_2(\tau))$. Further every genus 2 Heegaard decomposition of $E(K)$ is obtained in this manner; and two such Heegaard decompositions are isotopic (homeomorphic resp.), if and only if the corresponding tunnels are isotopic (homeomorphic resp.). Thus we sometimes represent an unknotting tunnel $\tau$ by a graph of the shape o— embedded in $V_2(\tau)$ as illustrated in Fig. 0.1.

In [BRZ], Boileau et al. completely classified unknotting tunnels for torus knots using the results on Nielsen equivalence classes of generator systems for

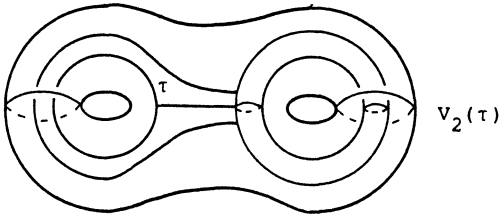



**Fig. 0.1**

Dedicated to Professor Yoko Tao for her 60th birthday

torus knot groups (see [C, Z]). For the relation between Heegaard splittings and the concept of the Nielsen equivalences in combinatorial group theory, see also [BCZ 1, 2, F, LM, Mh 1, 2]. Other criteria for distinguishing unknotting tunnels were given by [BM, K 3]. Bleiler and Moriah [BM] applied their method to distinguish the upper and lower tunnels for 2-bridge knots (cf. [F]); and Kobayashi [K 3] found other unknotting tunnels for two-bridge knots, and classified them up to homeomorphism.

The main purpose of this paper is to determine the non-simple knots in $S^3$ which admit unknotting tunnels, and to classify the unknotting tunnels for them (see 1.7, Theorems 2.1, 4.1, and 4.2). [A knot $K$ is said to be non-simple, if $E(K)$ contains an incompressible torus which is not boundary-parallel.] Roughly speaking, such knots are certain satellites of torus knots via 2-bridge links; and most of them admit precisely 4 (infinitely many resp.) unknotting tunnels up to homeomorphism (isotopy resp.).

In Sect. 1, we introduce concepts of a *dual tunnel* and a $(1,1)$-*tunnel*, which help us to understand geometrically the unknotting tunnels which we construct. We also review the unknotting tunnels for torus knots and 2-bridge knots from this view point. In Sect. 2, we determine the non-simple knots in $S^3$ admitting unknotting tunnels, and find all the unknotting tunnels for them. To do this we use the arguments of Kobayashi [K 1, 2] and the first author [Mt]. In Sect. 3, we introduce a very natural isotopy invariant for tunnels following Boyle [By], and discuss its relation to other invariants. We also recover the results mentioned in the above by using abelian reductions of the invariant (Examples 3.3 and 3.4). In Sect. 4, we classify the unknotting tunnels for non-simple knots in $S^3$ through the invariant. In the final section, we give the isotopy classification of the known unknotting tunnels for two-bridge knots through the invariant. These calculations show the invariant is very effective; and we conjecture that it is a complete invariant for isotopy types of unknotting tunnels.

The authors would like to thank Dr. Tsuyoshi Kobayashi for his enlightening suggestions and conversations.

## 1. Dual tunnels, (1, 1)-tunnels and examples

(1.1) *Dual tunnel.* Let $\tau$ be an unknotting tunnel for a knot $K$ in a closed orientable 3-manifold $M$, and recall the Heegaard decomposition $(V_1(\tau), V_2(\tau))$ of $E(K) = \mathrm{cl}(M - N(K))$. Suppose there exist a meridian disk $D$ of $V_1(\tau)$ and an annulus $A$ properly embedded in $V_2(\tau)$ such that:

(1) a component, say $\alpha$, of $\partial A$ is in $\partial V_2(\tau) - \partial E(K)$ and the other component $\partial A - \alpha$ is in $\partial E(K)$,

(2) $\alpha$ intersects $\partial D$ transversely in a single point.

Note that each of $V_1(\tau) \cup N(A; V_2(\tau))$ and $\mathrm{cl}(V_2(\tau) - N(A; V_2(\tau)))$ is a genus two handlebody. Let $c$ be a core of the solid torus $\mathrm{cl}(V_1(\tau) - N(D; V_1(\tau)))$, and let $\tau^*$ be a graph embedded in $V_1(\tau) \cup N(A; V_1(\tau))$ which is the union of $c$ and an arc $d$ joining $c$ and a point in $\partial E(K)$ (see Fig. 1.1). Then $\tau^*$ also is an unknotting tunnel for $K$; we call it a *dual tunnel* of $\tau$.

(1.2) $(1,1)$-*tunnel.* In the definition of a dual tunnel, if $\partial A - \alpha$ is a meridian of $K$, then the above decomposition of $E(K)$ induces a $(1,1)$-*decomposition* of $(M, K)$. Here a

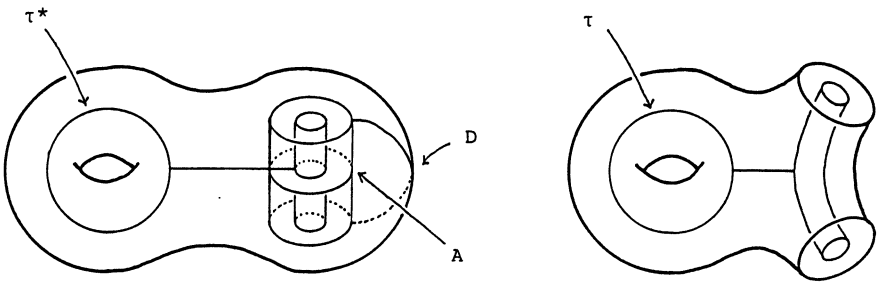



Fig. 1.1

$(g, b)$-*decomposition* of $(M, K)$ is a decomposition of $(M, K)$ into a union $(V_1, t_1) \cup (V_2, t_2)$, where $V_i$ is a genus $g$ handlebody and $t_i$ is a $b$-string trivial arc system in $V_i$ $(i=1, 2)$. In this case, we call $\tau$ and $\tau^*$ $(1,1)$-*tunnels* for $K$. Of course, every $(1,1)$-tunnel has a dual tunnel. Conversely, if an unknotting tunnel $\tau$ for $K \subset M$ has a dual tunnel, then it can be regarded as a $(1,1)$-tunnel for a knot $K'$ in a genus $\leqq 1$ 3-manifold $M'$ with $E(K') = E(K)$. In fact, $(M', K') = (E(K) \cup (S^1 \times D^2), S^1 \times 0)$, where a meridian of $S^1 \times D^2$ is identified with $\partial A - \alpha$, satisfies this property. We note that a knot which admits a $(1,1)$-decomposition has been called a 1-*bridge knot* in a lens space in [K 1] (cf. [Mt]). K. Taniyama pointed out the following proposition, which can be proved by using [Go, Theorem 1].

(1.3) **Proposition.** *Let $\tau$ be an arc in $M$ such that $\tau \cap K = \partial\tau$ and $\tau$ determines an unknotting tunnel for $K$. Let $\gamma$ and $\gamma'$ be two subarcs of $K$ cut off by $\partial\tau$. Then $\tau$ determines a $(1,1)$-tunnel for $K$ if and only if at least one of* $\mathrm{cl}(M - N(\tau \cup \gamma; M))$ *and* $\mathrm{cl}(M - N(\tau \cup \gamma'; M))$ *is a solid torus.*

In the following, we give some examples of $(1,1)$-tunnels and unknotting tunnels which do not have dual tunnels.

(1.4) *Unknotting tunnels for torus knots.* Let $K = T(p, q)$ be the nontrivial torus knot in $S^3$ of type $(p, q)$. Let $r$ and $s$ be integers such that $ps - qr = 1$. Then $E(K)$ is a Seifert fibred space $D(-r/p, s/q)$; that is, the Seifert fibred space whose base orbifold is a disk with two cone points, and the Seifert invariants of the singular fibres, say $u$ and $v$, are $-r/p$ and $s/q$ respectively (cf. [BRZ, Ms]). We see that $K$ has the following three unknotting tunnels $\tau_u$, $\tau_v$, and $\tau_0$: The tunnel $\tau_u$ ($\tau_v$ resp.) is obtained from the singular fibre $u$ ($v$ resp.) together with a lift of an arc in the base orbifold which joins a cone point and a point in the boundary. The tunnel $\tau_0$ is a lift of an essential arc properly embedded in the base orbifold (see Fig. 1.2).

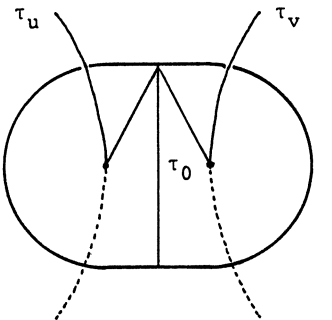



Fig. 1.2

Boileau et al. [BRZ] proved that unknotting tunnels for $K$ are those ones only up to isotropy, and classified them up to isotropy and homeomorphism (cf. [Mh 1]). Since $u$ and $v$ are cores of two solid tori which give a genus one Heegaard decomposition of $S^3$, we see that $\tau_u$ and $\tau_v$ are (1,1)-tunnels and are dual to each other. For $\tau_0$ we have:

(1.5) **Proposition.** *$\tau_0$ has a dual tunnel if and only if $p \equiv \pm 1 \bmod q$ or $q \equiv \pm 1 \bmod p$.*

*Proof.* If $p \equiv \pm 1 \bmod q$ or $q \equiv \pm 1 \bmod p$, then $\tau_0$ is isotopic to $\tau_u$ or $\tau_v$ (cf. [BRZ, Theorem 1]). Hence $\tau_0$ is a (1,1)-tunnel and has a dual tunnel. Suppose $\tau_0$ has a dual tunnel. Then as mentioned before, we can regard $\tau_0$ as a (1,1)-tunnel for a knot $K'$ in a genus $\leqq 1$ manifold $M'$ obtained from $S^3$ by Dehn surgery along $K$, where $K'$ is the image of $K$. Since $E(K) = E(K') = D(-r/p, s/q)$, $K'$ is a regular fibre of the Seifert fibration of $M'$ which is an extention of that of $E(K')$. In this case, (1,1)-decompositions for $K'$ are unique up to isotopy by [Mt, Theorem 3] (cf. [Mt, Corollary 4.1]). Thus, since we can regard $\tau_u$ and $\tau_v$ as (1,1)-tunnels for $K'$, $\tau_0$ is isotopic to $\tau_u$ or $\tau_v$. Hence we obtain the desired result by [BRZ, Theorem 1].

(1.6) *Unknotting tunnels for 2-bridge knots.* Let $K = S(\alpha, \beta)$ be a non-trivial 2-bridge knot of type $(\alpha, \beta)$ in $S^3$. Here $\alpha$ is an odd positive integer, and we may assume $\beta$ is even and $|\beta| < \alpha$. Then $\alpha/\beta$ is expanded into a unique continued fraction of the form

$$[2a_1, 2a_2, \ldots, 2a_{2g-1}, 2a_{2g}] = 2a_1 + (2a_2 + (\ldots + (2a_{2g-1} + (2a_{2g})^{-1})^{-1} \ldots)^{-1},$$

where $a_i$ is a non-zero integer ($1 \leqq i \leqq 2g$). Then $K$ has a genus $g$ Seifert surface, and it is illustrated as in Fig. 1.3.

Let $\tau_i$, $\varrho_i$, and $\varrho_i'$ ($i = 1, 2$) be the tunnels for $K$ as illustrated in Fig. 1.3. These are unknotting tunnels for $K$. $\tau_1$ and $\tau_2$ are called the upper and lower tunnels respectively, and the homeomorphism classification of the two tunnels were given by Funke [F] and Bleiler-Moriah [BM]. The remaining tunnels were found by Kobayashi [K 3], and he classified them up to homeomorphism. [It should be noted that $\varrho_i$ and $\varrho_i'$ ($i = 1, 2$) are mutually homeomorphic.]

Here we observe that these unknotting tunnels are all (1,1)-tunnels and that $\varrho_1$ and $\varrho_1'$ ($\varrho_2$ and $\varrho_2'$ resp.) are dual tunnels of $\tau_2$ ($\tau_1$ resp.). Let $\{(B_1, t_1), (B_2, t_2)\}$ be the 2-bridge decomposition of $(S^3, K)$ as in Fig. 1.3, where $(B_i, t_i)$ is a 2-string trivial tangle ($i = 1, 2$). Put $t_i = s_i \cup s_i'$ ($i = 1, 2$), then $\tau_i$ is an arc in $B_i$ joining $s_i$ and $s_i'$; and $\varrho_i$ ($\varrho_i'$ resp.) is an arc in $B_i$ such that $\varrho_i \cap t_i = \varrho_i \cap s_i = \partial \varrho_i$ ($\varrho_i' \cap t_i = \varrho_i' \cap s_i' = \partial \varrho_i'$ resp.) as

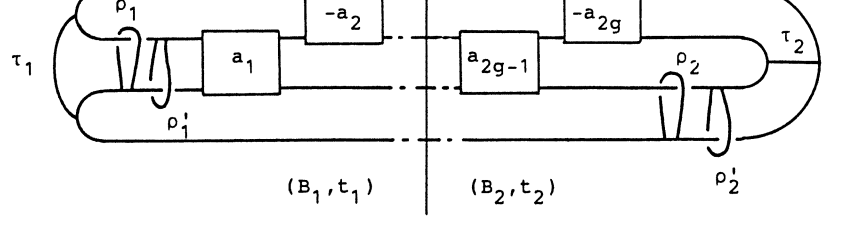



**Fig. 1.3**

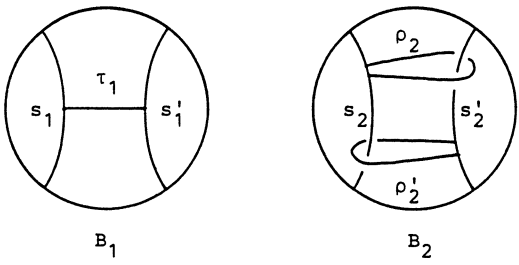



Fig. 1.4

illustrated in Fig. 1.4. Then both $B_1 \cup N(s_2'; B_2)$, say $V_1$, and $\mathrm{cl}(B_2 - N(s_2'; B_2))$, say $V_2$, are solid tori and $t_1 \cup s_2'$ ($s_2$ resp.) is a trivial arc properly embedded in $V_1$ ($V_2$ resp.). Moreover $\tau_1 \cup$(a subarc of $t_1 \cup s_2'$) [$\varrho_2 \cup$(a subarc of $s_2$) resp.] is a core of the solid torus $V_1$ ($V_2$ resp.). Thus $\tau_1$ and $\varrho_2$ are $(1,1)$-tunnels and are dual to each other. By using the solid tori $B_1 \cup N(s_2; B_2)$ and $\mathrm{cl}(B_2 - N(s_2; B_2))$, we see that $\varrho_2'$ is a $(1,1)$-tunnel and is a dual tunnel of $\tau_1$. Similarly we see that $\tau_2$, $\varrho_1$, and $\varrho_1'$ are $(1,1)$-tunnels and that $\varrho_1$ and $\varrho_1'$ are dual tunnels of $\tau_2$.

It should be noted that 2-bridge decompositions of 2-bridge knots and links are unique up to isotopy by Schubert [Sb], and hence the isotopy types of these tunnels do not depend on a particular choice of a 2-bridge decomposition. We conjecture that these are the only unknotting tunnels for 2-bridge knots up to isotopy. In Sect. 5, we give the isotopy classification of these tunnels.

(1.7) *Unknotting tunnels for non-simple knots.* First we construct a family of non-simple knots admitting unknotting tunnels. Let $K_0$ be a non-trivial torus knot $T(p,q)$ of type $(p,q)$ in $S^3$, and let $L = K_1 \cup K_2$ be a 2-bridge link $S(\alpha,\beta)$ of type $(\alpha,\beta)$ in $S^3$ with $\alpha \geqq 4$ (that is, $L$ is neither a trivial link nor a Hopf link). Since $K_2$ is a trivial knot, there is an orientation preserving homeomorphism $f: E(K_2) \to N(K_0)$ which takes a meridian $m_2 \subset \partial E(K_2)$ of $K_2$ to a fibre $h \subset \partial N(K_0) = \partial E(K_0)$ of the Seifert fibration $D(-r/p, s/q)$ of $E(K_0)$. We denote the knot $f(K_1) \subset N(K_0) \subset S^3$ by the symbol $K(\alpha,\beta;p,q)$.

The unknotting tunnels for the knot $K(\alpha,\beta;p,q)$ are described as follows. Let $\tau_1$ and $\tau_2$ be the upper and the lower tunnels for $L$ in $E(L)$. For $i=1,2$, put $T_i = \partial N(K_i) \subset \partial E(L)$, $a_i = \tau_1 \cap T_i$ and $b_i = \tau_2 \cap T_i$. We may assume that $a_2$ corresponds to the end point of each of the graphs $\tau_u$ and $\tau_v$ in the torus knot exterior $E(K_0)$. [Here we identify $E(K_2)$ with $N(K_0)$ via $f$.] Let $v_i$ be an arc in $T_i$ joining $a_i$ and $b_i$ ($i=1,2$), where $v_2$ is chosen to be a subarc of the boundary of the "cross section" of the Seifert fibred space $E(K_0) = D(-r/p,\ s/q)$ (see Fig. 1.5).

Put $\tau(1,u) = \tau_1 \cup \tau_u$, $\tau(1,v) = \tau_1 \cup \tau_v$, $\tau(2,u) = \tau_2 \cup v_2 \cup \tau_u$, and $\tau(2,v) = \tau_2 \cup v_2 \cup \tau_v$. Let $D$ be the Dehn twist homeomorphism of $E(K(\alpha,\beta;p,q))$ along the torus $\partial E(K_0) = \partial E(K_2)$ in the direction $l_2$, where $l_2$ is a prefered longitude in $\partial E(K_2)$ of $K_2$ [= the boundary of the "cross section" of $E(K_0)$]. For $i \in \{1,2\}$, $w \in \{u,v\}$ and an integer $n$, we denote $\tau(i,w,n) = D^n(\tau(i,w))$. Then these are really unknotting tunnels. In fact we have the following fact.

(1.8) **Proposition.** *For any $i \in \{1,2\}$, $w \in \{u,v\}$ and integer $n$, $\tau(i,w,n)$ is a $(1,1)$-tunnel for $K(\alpha,\beta;p,q)$. Moreover $\tau(1,u,n)$ ($\tau(1,v,n)$ resp.) and $\tau(2,v,n)$ ($\tau(2,u,n-1)$ resp.) are dual to each other.*

*Proof.* Let $\{(B_1,t_1),(B_2,t_2)\}$ be a 2-bridge decomposition of $(S^3,L)$. Put $V_i = \mathrm{cl}(B_i - N(t_i))$ ($i=1,2$), then $V_i$ is a genus two handlebody, and we may assume

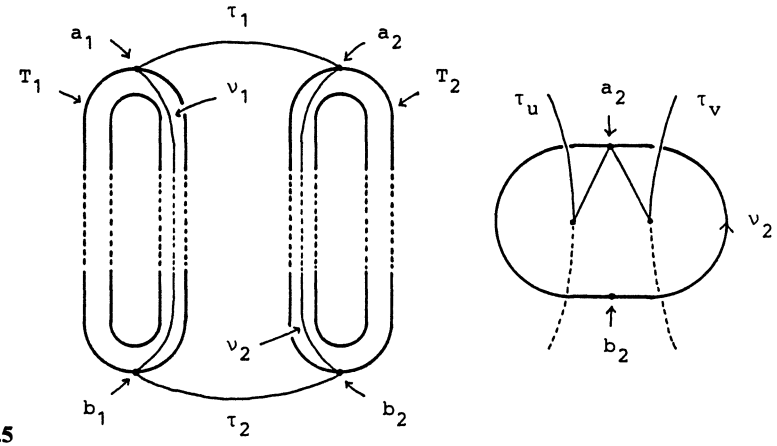



Fig. 1.5

that $\tau_i$ is an arc properly embedded in $V_i$. Let $m$ and $m'$ be two components of $\partial N(K_2)\cap\partial B_1$ $(=\partial N(K_2)\cap\partial B_2)$, then $m$ and $m'$ are meridians of $K_2$ and $f(m\cup m')$ are two fibres in $\partial E(K_0)$. Since $(m\cup m')\cap(a_2\cup b_2)=\emptyset$, $f(m\cup m')$ bounds an essential annulus, say $A$, properly embedded in $E(K_0)$ such that $A\cap\tau_u=\emptyset$ [see Fig. 1.6(a)]. Then $A$ decomposes $E(K_0)$ into two solid tori $U_1$ $(\supset\tau_u)$ and $U_2$, and $f(\partial N(K_2)\cap V_i)=\mathrm{cl}(\partial U_i-A)$ $(i=1,2)$. Put $W_i=U_i\cup_f V_i$ $(i=1,2)$. Then $W_1\cup W_2=E(K(\alpha,\beta;p,q))$, and this decomposition of $E(K(\alpha,\beta;p,q))$ shows that $\tau(1,u)$ and $\tau(2,v)$ are $(1,1)$-tunnels and are dual to each other(see Fig. 1.6(b)). Hence $\tau(1,u,n)$ and $\tau(2,v,n)$ are $(1,1)$-tunnels and are dual to each other for any integer $n$. Similarly we see that $\tau(1,v,n)$ and $\tau(2,u,n-1)$ are $(1,1)$-tunnels and are dual to each other for any integer $n$.

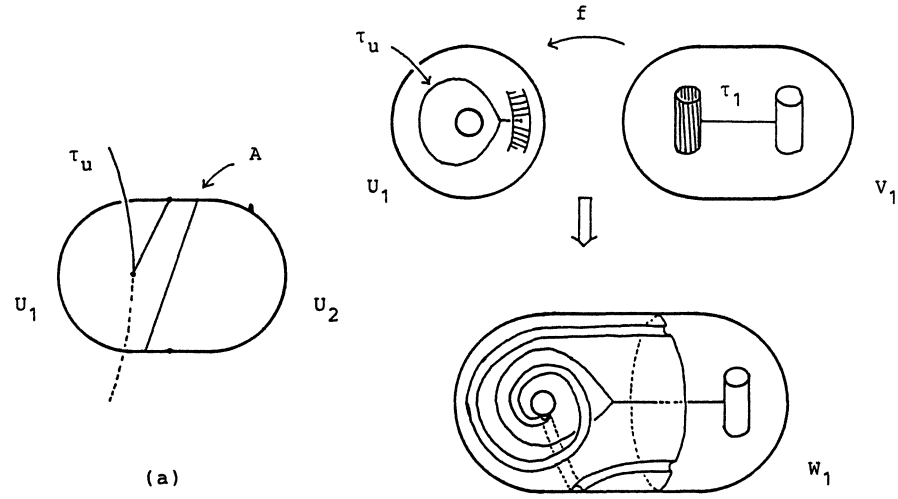



Fig. 1.6

In the next section, we prove that if a non-simple knot in $S^3$ admits an unknotting tunnel, then it is equivalent to some $K(\alpha, \beta; p, q)$, and the tunnels given in Proposition 1.8 are the only unknotting tunnels for $K(\alpha, \beta; p, q)$ up to isotopy. The isotopy and homeomorphism classifications of these tunnels are given in Sect. 4.

## 2. Determination of the non-simple knots which admit unknotting tunnels

(2.1) **Theorem.** *A non-simple knot in $S^3$ admits an unknotting tunnel, if and only if it is equivalent to a knot $K(\alpha, \beta; p, q)$. Further, any unknotting tunnel for $K(\alpha, \beta; p, q)$ is isotopic (homeomorphic resp.) to $\tau(i, w, n)$ ($\tau(i, w) = \tau(i, w, 0)$ resp.) for some $i \in \{1, 2\}$, $w \in \{u, v\}$ and an integer $n$.*

*Proof.* Since the "if" part of the first assertion is already proved by Proposition 1.8, we prove its "only if" part. Let $K$ be a non-simple knot in $S^3$ which admits an unknotting tunnel $\tau$. Let $(V_1, V_2) = (V_1(\tau), V_2(\tau))$ be the Heegaard decomposition of $E(K)$ determined by $\tau$ (cf. Introduction). If $E(K)$ is a Seifert fibred space, then by [Ms, Theorem 2] $K$ is a torus knot and is simple. Hence $E(K)$ admits a non-trivial torus decomposition (cf. [JS, Jo]). Then we have:

(2.2) **Lemma.** *Any family of tori which gives a non-trivial torus decomposition of $E(K)$ is ambient isotopic to a family $\Sigma$ such that $\Sigma_i = \Sigma \cap V_i$ ($i = 1, 2$) satisfies one of the following conclusions.*

(1) *$\Sigma_1$ consists of two disjoint non-separating essential annuli $A_1$ and $B_1$ in $V_1$ which satisfy the conclusion of* [K 1, *Lemma 3.4(i)*]. *In this case, $\Sigma_2$ consists of two mutually disjoint non-parallel essential annuli $A_2$ and $B_2$ in $V_2$ such that both of them are either* (a) *non-separating or* (b) *separating.*

(2) *$\Sigma_1$ consists of two mutually disjoint non-parallel separating essential annuli $A_1$ and $B_1$ in $V_1$ (cf.* [K 1, *Lemma 3.4*(ii)]*). In this case, $\Sigma_2$ consists of two mutually disjoint non-parallel essential annuli $A_2$ and $B_2$ in $V_2$ such that both of them are either* (a) *non-separating or* (b) *separating.*

(3) *$\Sigma_1$ consists of a single separating essential annulus $A_1$ in $V_1$. In this case, $\Sigma_2$ consists of a single separating essential annulus $A_2$ in $V_2$ such that it splits $V_2$ into* (a) *a genus two handlebody and the other or* (b) *a solid torus and the other.*

Since this lemma follows from the proof of [K 1, Theorem], we omitt the proof (cf. [K 2] and [Mt, Lemmas 1.1 and 1.5]). Note here that $E(K)$ contains no non-separating closed surface.

In the following, we show that the conclusions (1) and (2) of Lemma 2.2 do not occur and that the conclusion (3) implies the desired result.

*Case (1-a).* $\Sigma$ satisfies the conclusion (1-a) of Lemma 2.2. By the argument of [K 1, p. 452, Case 2.2.1], $E(K)$ is a union of a non-trivial 2-bridge knot exterior in $S^3$ and a Seifert fibred space over a Möbius band with one hole with at most one singular fibre. Then $E(K)$ contains a Klein bottle. Thus we have a contradiction and Case (1-a) does not occur.

*Case (1-b).* $\Sigma$ satisfies the conclusion (1-b) of Lemma 2.2. By the argument of [K 1, p. 452, Case 2.2.2], $E(K)$ is a union of a non-trivial 2-bridge knot exterior in $S^3$, say $M_1$, and a Seifert fibred space over an annulus with one or two singular fibres, say $M_2$. If a fibre of $M_2$ in $\partial E(K)$ is a meridian of $K$, then by [O, p. 19, Remark (ii)] $S^3$

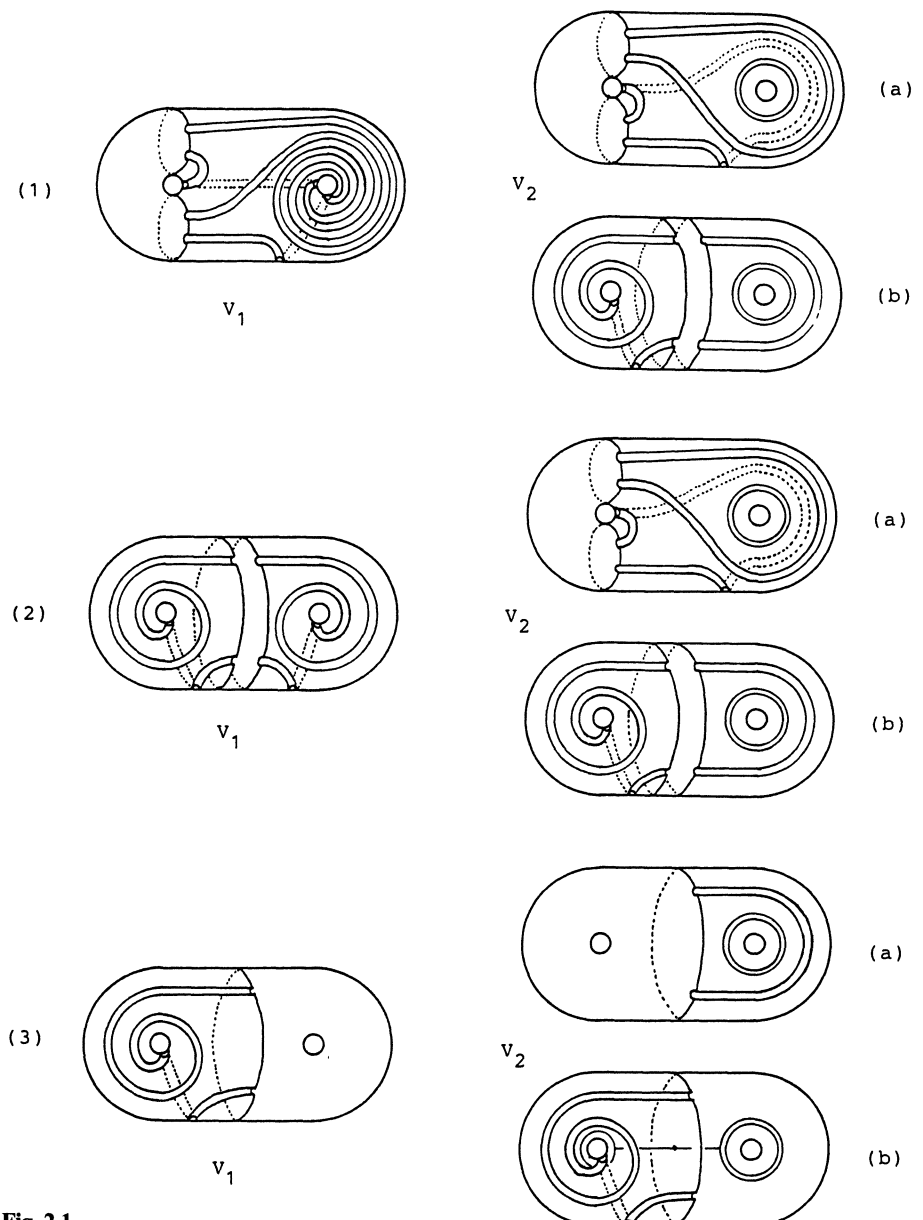



**Fig. 2.1**

must have a lens space summand. This is a contradiction. Hence $M_2 \cup N(K)$, say $M_3$, is a Seifert fibred space over a disk with one, two or three singular fibres. Since $S^3$ $(=M_1 \cup M_3)$ contains no incompressible torus, $M_3$ has one singular fibre. Since a fibre in $\partial M_3$ is glued to a meridian in $\partial M_1$ of the 2-bridge knot (cf. [K 1, p. 452, Case 2.2.2]), a meridian in $\partial M_3$ as a solid torus intersects a meridian in $\partial M_1$ in more than one point. This is a contradiction because every non-trivial 2-bridge knot has Property $P$ by [Ts]. Hence Case (1-b) does not occur.

*Case (2-a).* $\Sigma$ satisfies the conclusion (2-a) of Lemma 2.2. By the argument of [K 1, p. 452, Case 2.2.2], $E(K)$ is a union of a non-trivial 2-bridge knot exterior in $S^3$, and a Seifert fibred space over an annulus with one or two singular fibres. Since $A_1$ and $B_1$ are essential, we see the Seifert fibred space has two singular fibres. This is a contradiction, since $S^3$ has no essential torus. So Case (2-a) does not occur.

*Case (2-b).* $\Sigma$ satisfies the conclusion (2-b) of Lemma 2.2. We may assume that $B_2$ splits $V_2$ into a genus two handlebody and the other, and that $\partial B_2$ is glued to $\partial B_1$. Put $T=B_1\cup B_2$. Then we see that $T$ splits $E(K)$ into a knot exterior in some genus $\leqq 1$ 3-manifold, say $M_1$, and a Seifert fibred space over an annulus with one singular fibre, say $M_2$. Note that $\partial B_1=\partial B_2\subset\partial M_1$ is a meridian of the knot in $M_1$. Put $M_3=M_2\cup N(K)$, then by the argument of Case (1-b), $M_3$ is a Seifert fibred space over a disk with one singular fibre, and a meridian in $\partial M_3$ as a solid torus intersects $\partial B_1$ in more than one point. Since $M_1\cup M_3=S^3$, by Cyclic Surgery Theorem of [CGLS], $M_1$ is a Seifert fibred space. Since $M_1\subset S^3$, by [Ms, Theorem 2], the base orbifold of $M_1$ is a disk with two cone points. Now recall that the incompressible torus $A_1\cup A_2$ is contained in $M_1$ and that $\partial M_1=B_1\cup B_2$. This means that $A_1\cup A_2$ is parallel to $B_1\cup B_2$. This is a contradiction and Case (2-b) does not occur.

*Case (3-a).* $\Sigma$ satisfies the conclusion (3-a) of Lemma 2.2. By using Cyclic Surgery Theorem of [CGLS] as in Case (2-b), we see that $\Sigma$ splits $E(K)$ into a Seifert fibred space over a disk with two singular fibres, say $M_1$, and a Seifert fibred space over an annulus with one singular fibre, say $M_2$. Put $G=M_1\cap\partial V_1\ (=M_1\cap\partial V_2)$. Then $G$ is a torus with two holes properly embedded in $M_1$, and $G$ naturally extends to a torus which gives a genus one Heegaard decomposition of a 3-manifold, say $M$, which is obtained from $M_1$ and a solid torus, say $V$, by glueing a meridian of $V$ to a component of $\partial G$. Since $M$ has a genus one Heegaard splitting, the core of $V$ is a regular fibre of the Seifert fibration of $M$ which is an extension of that of $M_1$. Hence by [Mt, Theorem 3], for $i=1,2$ there exists a disk $\Delta_i$ in $V_i$ such that $\Delta_i\cap A_i=\partial\Delta_i\cap A_i=\alpha_i$ is a single essential arc properly embedded in $A_i$, $\mathrm{cl}(\partial\Delta_i-\alpha_i)=\beta_i$ is a single essential arc properly embedded in $G\subset\partial V_i$, and $\beta_1\cap\beta_2=\emptyset$. Since $\beta_1\cap\beta_2=\emptyset$, by performing isotopies of type A (for the definition, see [Ja, Ch. II]) along $\Delta_1$ and $\Delta_2$ simultaneously, we see that $\Sigma$ is ambient isotopic to $\Sigma'$ satisfying the conclusion (3-b) of Lemma 2.2. Hence this case is contained in the next case.

*Case (3-b).* $\Sigma$ satisfies the conclusion (3-b) of Lemma 2.2. Let $X_i$ and $Y_i$ be the closure of each component of $V_i-A_i$ $(i=1,2)$, where $X_i$ is a solid torus. Put $M_1=X_1\cup X_2$ and $M_2=Y_1\cup Y_2$. Then $M_1$ is a Seifert fibred space over a disk with two singular fibres. Hence $M_1=E(K_0)$ for some non-trivial torus knot $K_0$ in $S^3$. Put $M_3=M_2\cup N(K)$. Then by [K 1, Lemma 5.1], $M_3$ is an exterior of a $(1,1)$-bridge knot (cf. 1.2). That is, there is a knot $K'$ in a closed orientable 3-manifold $M'$, such that (i) $M_3=E(K')$ and (ii) $(M',K')\cong(W_1,t_1)\cup(W_2,t_2)$ where $W_i$ is a solid torus and $t_i$ is a trivial arc in $W_i$ for $i=1,2$. Here each component of $\partial A_1=\partial A_2$ is a meridian of $K'$; $K$ is a core of $W_2$, and there exists a disk $\Delta$ in $W_2$ such that $\partial\Delta=(\text{an arc in }\partial W_2)\cup(K'\cap W_2)$ and $\Delta\cap K=\emptyset$ (see Fig. 2.2). On the other hand, since $\mathrm{cl}(S^3-M_3)=M_1$ is not a solid torus, $M_3$ is a solid torus (cf. [R, p. 107]). Since each component of $\partial A_1=\partial A_2$ is a fibre of the Seifert fibred space $M_1=E(K_0)$, it is a longitude of the solid torus $M_3$. Hence $(M',K')\cong(S^3,$ a trivial knot). Let $L$ be the link $K\cup K'$ in $M'\cong S^3$. Now the following lemma completes the proof of the first assertion of Theorem 2.1.

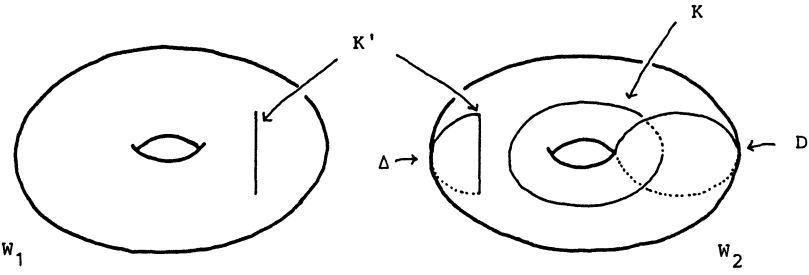



**Fig. 2.2**

(2.3) **Lemma.** *$L$ is a 2-bridge link.*

*Proof.* Let $D$ be a meridian disk of $W_2$ as illustrated in Fig. 2.2 and $N(D)$ a regular neighborhood of $D$ in $W_2$, such that $N(D)\cap K$ is a single arc and $\Delta\cap N(D)=\emptyset$. Put $B_1=W_1\cup N(D)$ and $B_2=\mathrm{cl}(W_2-N(D))$, then $B_i$ is a 3-ball ($i=1,2$). Put $B_i\cap K=\alpha_i$ and $B_i\cap K'=\beta_i$ ($i=1,2$). Then by the existence of $\Delta$ in $B_2$, $(B_2,\alpha_2\cup\beta_2)$ is a 2-string trivial tangle. On the other hand, we see (i) $\pi_1(B_1-(\alpha_1\cup\beta_1))\cong Z*Z$, (ii) $\pi_1(B_1-\alpha_1)\cong Z$, and (iii) $\pi_1(B_1-\beta_1)\cong Z$. In fact, these follow from the facts that (i) $\beta_1$ is a trivial arc in $V_1$, (ii) $\mathrm{cl}(B_1-N(D))=W_1$, and (iii) $\beta_1\cup\beta_2=K'$ is a trivial knot in $S^3$ ($=B_1\cup B_2$) and $\beta_2$ is a trivial arc in $B_2$, and therefore $\beta_1$ is a trivial arc in $B_1$. Hence by [Go, Theorem 1] (cf. [Tc, Theorem]), $(B_1,\alpha_1\cup\beta_1)$ is a 2-string trivial tangle, and $L$ is a 2-bridge link in $S^3$.

Finally, we prove the second assertion of Theorem 2.1. Let $\tau$ be an unknotting tunnel for $K=K(\alpha,\beta;p,q)$. Then by the preceding arguments, we may assume that the Heegaard decomposition $(V_1,V_2)=(V_1(\tau),V_2(\tau))$ of $E(K)$ satisfies the assumption of Case (3-b). Thus as a graph of the shape o— (cf. Introduction), $\tau$ is divided into a subarc $\tau'$ and the graph $\tau''=\mathrm{cl}(\tau-\tau')$ by the annulus $A_2$ [see Figs. 2.1(3-b) and 1.6(b)]. By the proof of Lemma 2.3 and the uniqueness of 2-bridge decompositions of a 2-bridge link (see [Sb]), we may assume that $\tau'=\tau_1$ or $\tau_2$. Since any essential annulus properly embedded in $M_1$ is isotopic to an annulus saturated in the Seifert fibration of $M_1$, $\tau''$ is isotopic to one of $\tau_u$, $\tau_v$, $\nu_2\cup\tau_u$, and $\nu_2\cup\tau_v$ by an ambient isotopy of $M_1$. Hence by noting Dehn twist homeomorphism along $\partial M_1$, we see that $\tau$ is isotopic to some $\tau(i,w,n)$ and that $\tau$ is homeomorphic to some $\tau(i,w)$.

(2.4) *Remark.* (1) Kobayashi informed us that he has an alternative proof of the fact that any unknotting tunnel for a torus knot is isotopic to one of $\tau_u$, $\tau_v$, and $\tau_0$ (cf. [BRZ]) through the isotopy of type A arguments.

(2) As a corollary of Theorem 2.1, we see that knots which admit unknotting tunnels are prime. This has already been proved by Norwood [N] and Scharlemann [Sl].

## 3. Invariants for tunnels

Let $K$ be a knot in a closed orientable 3-manifold $M$, and $p$ a point in $\partial E(K)$. Let $G(K)=\pi_1(E(K),p)$ be the knot group, and

$$\mathscr{P}(K)=\mathrm{Im}[j_*:\pi_1(\partial E(K),p)\to\pi_1(E(K),p)]$$

be its peripheral subgroup. Let $\tau$ be a tunnel for $K$. Choose paths $\alpha$ and $\beta$ on $\partial E(K)$ joining $p$ and each of the endpoints of $\tau$, and let $\lambda(\tau)$ be an element of $G(K)$ represented by the loop $\alpha \cup \tau \cup \beta$ with an arbitral orientation. Then the union of the double cosets of $\mathscr{P}(K)$ in $G(K)$ represented by $\lambda(\tau)$ and $\lambda(\tau)^{-1}$ does not depend on the choices of the paths $\alpha$ and $\beta$ and the orientation of the loop $\alpha \cup \tau \cup \beta$ (cf. [By, Lemma 1]). We denote it by the symbol $\Lambda(\tau)$. This is an invariant for the isotopy class (in fact homotopy class) of the tunnel $\tau$.

Suppose $M$ is a homology sphere. Then this invariant has the following abelian reductions. Let $A_\infty(K)$ be the knot module of $K$; that is the $Z\langle t\rangle$-module $H_1(E_\infty(K))$, where $Z\langle t\rangle$ is the integral group ring of the infinite cyclic group $\langle t\rangle$ generated by the symbol $t$, and $E_\infty(K)$ is the infinite cyclic cover of $E(K)$. Let $\tilde{\tau}$ be a lift of a tunnel $\tau$ to $E_\infty(K)$. Then it determines an element of $H_1(E_\infty(K), \partial E_\infty(K)) \cong H_1(E_\infty(K))$; and this is an invariant of the isotopy class of $\tau$ modulo multiplication by $\pm t^i$. We denote it by the symbol $\lambda_\infty(\tau)$. Recall that $A_\infty(K)$ is isomorphic to the commutator quotient $G'(K)/G''(K)$. Then we see that $\lambda_\infty(\tau)$ is obtained from $\Lambda(\tau)$ by choosing a representative $\lambda(\tau)$ which is contained in $G'(K)$, and considering the element of $A_\infty(K)$ determined by it.

For a natural number $n$, let $A_n(K)$ be the $Z\langle t\rangle$-module $A_\infty(K)/(t^n-1)A_\infty(K)$. This is isomorphic to the first homology group of the $n$-fold cyclic cover $\Sigma_n(K)$ of $S^3$ branched over $K$. Let $\lambda_n(\tau)$ be the element of $A_n(K)$ represented by $\lambda_\infty(\tau)$. Then this is an invariant of $\tau$ modulo multiplication by $\pm t^i$. In particular $\lambda_2(\tau)$ is an invariant of $\tau$ modulo multiplication by $\pm 1$. The following proposition gives a condition for a tunnel to be an unknotting tunnel.

(3.1) **Proposition.** *If $\tau$ is an unknotting tunnel, then $\lambda_n(\tau)$ is a generator of the $Z\langle t\rangle$-module $A_n(K)$ for each $n \in N \cup \{\infty\}$.*

Next, we state without proof a relation between $\lambda_\infty(\tau)$ and $\lambda_\infty(\tau^*)$, where $\tau^*$ is a dual tunnel of $\tau$, in terms of the invariant introduced by Lustig and Moriah [LM]. Let $M(K,0)$ be the 3-manifold obtained from $S^3$ by performing 0-surgery on $K$. For an unknotting tunnel $\tau$, put $\hat{V}_2(\tau) = \mathrm{cl}(M(K,0) - V_1(\tau))$. Then $(V_1(\tau), \hat{V}_2(\tau))$ gives a genus two Heegaard decomposition of $M(K,0)$. Let $\{x, y\}$ and $\{u, v\}$ be generator systems of $\pi_1(V_1(\tau))$ and $\pi_1(\hat{V}_2(\tau))$ respectively. The elements of the free group $\langle x, y\rangle$ representing the same elements of $\pi_1(M(K,0))$ as $u$ and $v$ are also denoted by the symbols $u$ and $v$ respectively. Let $\phi$ be the composition of the natural homomorphisms $\langle x, y\rangle \to \pi_1(M(K;0)) \to \langle t\rangle$, and $\delta(\tau) \in Z\langle t\rangle$ be the determinant of the matrix $\begin{pmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{pmatrix}^{\phi}$. Here $\partial/\partial x$ and $\partial/\partial y$ denote Fox's free derivatives. The arguments of [LM] and [FK] shows that it is an invariant of the isotopy class of the ordered Heegaard splitting modulo the ideal generated by the Alexander polynomial of $K$ and multiplication by $\pm t^i$. Then we have:

(3.2) **Proposition.** *Suppose $\tau$ has a dual tunnel, and the loop $\alpha$ in 1.1 represents the homology class $p \in H_1(E(K)) \cong Z$. Then*

$$\lambda_\infty(\tau) \doteq \varrho_p(t)\delta(\tau)\lambda_\infty(\tau^*),$$

*where $\varrho_p(t) = (t^p-1)/(t-1)$.*

We now give some applications of these abelian reductions of the invariant.

(3.3) *Example.* Let $K = T(p, q)$ be a torus knot. Then $A_\infty(K) \cong Z\langle t\rangle/\Delta(t)$, where

$$\Delta(t) = (t^{pq}-1)(t-1)/(t^p-1)(t^q-1),$$

and we obtain the following identities in $A_\infty(K)$ by using Fox's free differential calculus (cf. [BZ, Example 9.15]):

$$\lambda_\infty(\tau_u) \doteq \varrho_s(t^p), \quad \lambda_\infty(\tau_v) \doteq \varrho_r(t^q), \quad \lambda_\infty(\tau_0) \doteq \varrho_s(t^p)\varrho_r(t^q).$$

Here $r$ and $s$ are integers such that $ps-qr=1$.

From these identities, we can recover the discrimination up to isotopy of the unknotting tunnels for torus knots (see [BRZ, Theorem 1]). The homeomorphism classification is identical with the isotopy classification, since the mapping class group of $E(K)$ is the cyclic group of order 2 generated by an involution which preserves each of the unknotting tunnels (cf. [Sr]).

(3.4) *Example.* Let $K=S(\alpha,\beta)$ be a 2-bridge knot. Then $H_1(\Sigma_2(K)) \cong Z/\alpha Z$, and we have the following identities in $H_1(\Sigma_2(K))$:

$$\lambda_2(\tau_1) \equiv \pm 2^{-1}, \quad \lambda_2(\tau_2) \equiv \pm 2^{-1}\beta^{-1},$$
$$\lambda_2(\varrho_1) \equiv \lambda_2(\varrho_1') \equiv \pm 1, \quad \lambda_2(\varrho_2) \equiv \lambda_2(\varrho_2') \equiv \pm \beta^{-1}.$$

*Proof.* Recall the 2-bridge decomposition $(S^3, K)=(B_1, t_1)\cup(B_2, t_2)$ as illustrated in Fig. 1.3. The double cover $\tilde{B}_i$ of $B_i$ branched over $t_i$ is a solid torus for $i=1,2$, and the double cover $\Sigma_2(K)$ of $S^3$ branched over $K$ is obtained from $\tilde{B}_i$ $(i=1,2)$ by identifying their boundaries according to the identity

$$\begin{pmatrix} m_2 \\ l_2 \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} l_1 \\ m_1 \end{pmatrix}, \tag{3.5}$$

where $l_i$ and $m_i$ are the longitude and the meridian of the solid torus $\tilde{B}_i$ $(i=1,2)$, and $\gamma$ and $\delta$ are integers such that $\alpha\delta-\beta\gamma=1$. Thus we have $H_1(\Sigma_2(K)) \cong Z/\alpha Z$, where the homology classes $[l_1]$ and $[l_2]$ corresponds to 1 and $\gamma \equiv -\beta^{-1}$ respectively. Note that $H_1(\Sigma_2(K)) \cong H_1(\Sigma_2(K), \tilde{K})$ where $\tilde{K}$ is a lift of $K$ to $\Sigma_2(K)$. Thus for each tunnel $\tau$, $\lambda_2(\tau)$ is identified with the homology class in $H_1(\Sigma_2(K), \tilde{K})$ represented by a lift $\tilde{\tau}$ of $\tau$. [Here we regard $\tau$ as an arc in $S^3$ such that $\tau \cap K = \partial\tau$.] From Fig. 3.1, we see $\lambda_2(\varrho_1)=\lambda_2(\varrho_1')=\pm[l_1]$, and $(1-t)\lambda_2(\tau_1)=\pm[l_1]$. Since $t=-\mathrm{id}$, we have $\lambda_2(\tau_1) = \pm 2^{-1}[l_1]$. Similarly, we have $\lambda_2(\varrho_2)=\lambda_2(\varrho_2')=\pm[l_2]=\beta^{-1}[l_1]$, and $\lambda_2(\tau_2) = 2^{-1}[l_2]=\pm\beta^{-1}[l_1]$.

The above identities enable us to distinguish the tunnels $\{\tau_i, \varrho_i\ (i=1,2)\}$ up to isotopy, and together with Schubert's result [Sb], we can recover the homeomorphism classification given by [BM, K 3] of these tunnels (see Sect. 5).

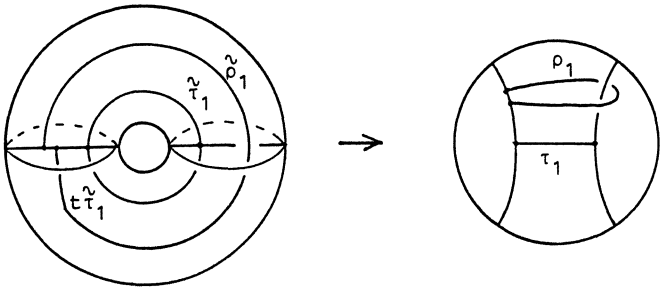



**Fig. 3.1**

However, we cannot distinguish the tunnels $\varrho_i$ and $\varrho_i'$ $(i=1,2)$ up to isotopy by $\lambda_2$-invariant. In Sect. 5, we do this by using $\Lambda$-invariant.

In the next section, we classify the unknotting tunnels for non-simple knots listed by Theorem 2.1 through $\Lambda$-invariant.

## 4. Classification of the unknotting tunnels for $K(\alpha, \beta; p, q)$

In this section, we prove the following theorems, which give the isotopy and homeomorphism classifications of the unknotting tunnels for the knot $K=K(\alpha, \beta; p, q)$ given by Theorem 2.1 (cf. 1.7).

(4.1) **Theorem.** *For $i, i' \in \{1,2\}$, $x \in \{u, v\}$, and integers $n$ and $n'$, we have the following:*

(1) *$\tau(i, x, n)$ is isotopic to $\tau(i, x, n')$, if and only if either $n=n'$ or $\beta \equiv \pm 1 \bmod \alpha$.*

(2) *$\tau(1, x, n)$ is isotopic to $\tau(2, x, n')$, if and only if $\beta \equiv \pm 1 \bmod \alpha$.*

(3) *$\tau(i, u, n)$ is not isotopic to $\tau(i', v, n')$.*

(4.2) **Theorem.** *For $i, i' \in \{1,2\}$ and $x \in \{u, v\}$, we have the following:*

(1) *$\tau(1, x)$ is homeomorphic to $\tau(2, x)$, if and only if $\beta^2 \equiv 1 \bmod \alpha$.*

(2) *$\tau(i, u)$ is not homeomorphic to $\tau(i', v)$.*

Hence we obtain the following table which gives the number of isotopy and homeomorphism classes of unknotting tunnels for $K(\alpha, \beta; p, q)$.

(4.3) **Table**

| | Isotopy class | Homeomorphism class |
|---|---|---|
| $\beta \not\equiv \pm 1 \bmod \alpha$ | $\infty$ | 4 if $\beta^2 \not\equiv 1 \bmod \alpha$<br>2 if $\beta^2 \equiv 1 \bmod \alpha$ |
| $\beta \equiv \pm 1 \bmod \alpha$ | 2 | 2 |

*Proof of Theorem 4.1.* The "if" part can be seen by using the Seifert fibred structure of $E(S(\alpha, \pm 1))$ [cf. (4.15)]. So we prove the "only if" part. Recall the notations introduced in 1.7. Put $G(L)=\pi_1(E(L), a_1)$, and let $\mu$ be an element of $G(L)$ represented by the loop based at $a_1$ which runs $v_1$, $\tau_2$, $v_2$, and $\tau_1$ in this order. The groups $\pi_1(T_1, a_1)$ and $\pi_1(T_2, a_2) \cong \pi_1(T_2 \cup \tau_1, a_1)$ can be considered as subgroups of $G(L)$; we denote them by the symbols $\mathscr{P}_1$ and $\mathscr{P}_2$ respectively.

(4.4) **Lemma.** (1) *The subset $\mathscr{P}_1 \mu \mathscr{P}_2$ of $G(L)$ contains the identity, if and only if $\beta \equiv \pm 1 \bmod \alpha$.*

(2) *A conjugate of $\mathscr{P}_1$ and a conjugate of $\mathscr{P}_2$ has a nontrivial intersection, if and only if $\beta \equiv \pm 1 \bmod \alpha$.*

(3) *Suppose $\beta \equiv \pm 1 \bmod \alpha$. Then the intersection of a conjugate of $\mathscr{P}_1$ and a conjugate of $\mathscr{P}_2$ is generated by the element represented by a regular fibre of the Seifert fibration of $E(L)$.*

*Proof.* (1) Suppose $\mathscr{P}_1 \mu \mathscr{P}_2$ contains the identity element. Then there is a continuous map $g$ from $D=[1,2] \times [1,2]$ to $E(L)$ such that $g(\{i\} \times [1,2]) \subset T_i$ $(i=1,2)$ and

$g([1,2]\times\{i\})=\tau_i$ $(i=1,2)$. From this map, we can constract a continuous map $\hat{g}$ from $D$ to $S^3$, which satisfies the following conditions.

(i) $\hat{g}^{-1}(K_i)=\{i\}\times[1,2]$ $(i=1,2)$.

(ii) $\hat{g}([1,2]\times\{i\})=\hat{\tau}_i$ $(i=1,2)$, where $\hat{\tau}_i$ is an arc in $S^3$ such that $\hat{\tau}_i\cap L=\partial\hat{\tau}_i$ obtained from $\tau_i$ by a radial extension.

As in the proof of Example 3.4, we consider the double cover $\Sigma_2(L)$ of $S^3$ branched over $L$, which is obtained from the two solid tori $\tilde{B}_i$ $(i=1,2)$ by identifying their boundaries as indicated by (3.5). Since $D$ is contractible, $g$ has a lift to the double cover of $E(L)$; so $\hat{g}$ has a lift $\tilde{g}: D\to\Sigma_2(L)$. Let $[\tilde{D}]$ denote the 2-chain $\tilde{g}_*[D]$ in $\Sigma_2(L)$. Then as in the proof of Example 3.4, we see $\partial([\tilde{D}]-t[\tilde{D}])$ is homologous to $[l_1]\pm[l_2]$. Hence we have

$$0=[l_1]\pm[l_2]=(1\pm\beta^{-1})[l_1] \quad \text{in} \quad H_1(\Sigma_2(L))=\langle[l_1]\,|\,\alpha[l_1]=0\rangle.$$

Thus we see $\beta\equiv\pm1 \bmod\alpha$.

(2) Suppose the assumption of (2) is satisfied. Then there is a non-degenerate map from $(S^1\times I, S^1\times\partial I)$ to $(E(L),\partial E(L))$, and hence $E(L)$ has an essential annulus by Annulus Theorem (cf. [Ja, VIII.13]). Since $E(L)$ is simple, $E(L)$ is a Seifert fibred space and hence $\beta\equiv\pm1 \bmod\alpha$.

(3) Suppose $\beta\equiv\pm1 \bmod\alpha$, and let $f$ be an element of $G(L)$ represented by a regular fibre. Then

$$G(L)/\langle f\rangle\cong\langle x_1,x_2\,|\,(x_1x_2)^{\alpha/2}=1\rangle,$$

where $x_i$ generates $\mathscr{P}_i/\langle f\rangle$ $(i=1,2)$. Noting that this group is isomorphic to $Z*Z_{\alpha/2}$ and $\alpha/2\geqq2$, we obtain the desired result.

Recall that the torus knot group $\pi_1(E(K_0))$ has the presentation $\langle u,v\,|\,u^pv^{-q}=1\rangle$, where $u$ and $v$ are represented by the singular fibres, and $\pi_1(\partial E(K_0))$ is the free Abelian group generated by the meridian $m_0=u^rv^{-s}$ and the regular fiber $h=u^p=v^q$. We may assume $p>q\geqq2$.

(4.5) **Lemma.** (1) *For integers $a$, $b$, $c$, and $d$, let $w(a,b,c,d)=(m_0^ah^b)u(m_0^ch^d)\in G(K_0)$. Then the followings are the only solutions of the equation $w(a,b,c,d)=u^{\pm1}$ or $v^{\pm1}$:*

(i) $w(0,b,0,-b)=u$.

(ii) $w(-1,b,0,-b)=v$ *when* $q=p-1$.

(iii) $w(-1,b-1,-1,-b)=\bar{u}$ *when* $(p,q)=(3,2)$.

(2) *For integers $a$, $b$, $c$, and $d$, let $w'(a,b,c,d)=(m_0^ah^b)v(m_0^ch^d)\in G(K_0)$. Then the followings are the only solutions of the equation $w'(a,b,c,d)=u^{\pm1}$ or $v^{\pm1}$:*

(i) $w'(0,b,0,-b)=v$.

(ii) $w'(1,b,0,-b)=u$ *when* $q=p-1$.

*Proof.* Consider the quotient group

$$G(K_0)/\langle h\rangle\cong\langle u\,|\,u^p=1\rangle*\langle v\,|\,v^q=1\rangle,$$

and consider its elements $\hat{w}(a,c)=(u^r\bar{v}^s)^au(u^r\bar{v}^s)^c$ and $\hat{w}'(a,c)=(u^r\bar{v}^s)^av(u^r\bar{v}^s)^c$. Since $G(K_0)/\langle h\rangle$ is the free product, the length $|w|$ of an element $w$ is defined (cf. [LS, p. 176]). Now Lemma 4.5 follows immediately from the following sublemma.

(4.6) **Sublemma.** (1) $\hat{w}(a,c)$ *has length* 1, *if and only if it is of one of the following forms:*

(i) $\hat{w}(0,0)=u$.

(ii) $\hat{w}(-1,0)=v$ *when* $q=p-1$.

(iii) $\hat{w}(-1,-1)=\bar{u}$ *when* $(p,q)=(3,2)$.

(2) *$\hat{w}'(a,c)$ has length 1, if and only if it is of one of the following form.*
(i) $\hat{w}'(0,0)=u$.
(ii) $\hat{w}'(1,0)=u^r$ *when* $p-1\equiv 0 \bmod q$.

*Proof of Sublemma 4.6.* This follows from a case by case argument noting the assumption that $p>q\geqq 2$ and $ps-qr=1$. We describe only the proof of (1) for the case $a\leqq -1$. Suppose $a\leqq -1$. Then if $c\geqq 1$, we have $\hat{w}(a,c)=(v^s\bar{u}^r)^{|a|-1}v^su\bar{v}^s(u^r\bar{v}^s)^{c-1}$, and hence $|\hat{w}(a,c)|>1$. Suppose $c=0$. Then $\hat{w}(a,0)=(v^s\bar{u}^r)^{|a|-1}v^su^{-r+1}$; so $|\hat{w}(a,0)|=1$ if and only if $a=-1$ and $-r+1\equiv 0 \bmod p$. Since $r\equiv -q^{-1} \bmod p$, we have $q+1\equiv 0 \bmod p$. Since $p>q\geqq 2$, this is equivalent to the condition that $q+1=p$. Thus $s\equiv p^{-1}\equiv 1 \bmod q$, and hence $\hat{w}(-1,0)=v^s=v$. Suppose $c\leqq -1$. Then

$$\hat{w}(a,c)=(v^s\bar{u}^r)^{|a|-1}v^su^{-r+1}v^su^{-r}(v^su^{-r})^{|c|-1}.$$

If $|\hat{w}(a,c)|=1$, we have $-r+1\equiv 0 \bmod p$ and $2s=0 \bmod q$. Thus we have $q+1\equiv 0 \bmod p$ and $2\equiv 0 \bmod q$. So we have $(p,q)=(3,2)$ since $p>q\geqq 2$. Then $\hat{w}(a,c)=(v\bar{u})^{|a|-1}\bar{u}(v\bar{u})^{|c|-1}$, and we see $|\hat{w}(a,c)|=1$ if and only if $a=c=-1$. The other cases can be treated similarly.

*Proof of Theorem 4.1* (continued). Put $G=\pi_1(E(K(\alpha,\beta;p,q)),a_1)$. Then $G(L)=\pi_1(E(L),a_1)$ and $G(K_0)=\pi_1(E(K_0),a_2)\cong\pi_1(E(K_0)\cup\tau_1,a_1)$ can be identified with subgroups of $G$; and $G$ is the free product of $G(L)$ and $G(K_0)$ with the amalgamated subgroup $\mathscr{P}_2=\pi_1(T_2\cup\tau_1,a_1)$. Here the meridian $m_2$ and the longitude $l_2$ of $K_2$ are identified with the element $h$ $(=u^p=v^q)$ and $u^r\bar{v}^s$ of $G(K_0)=\langle u,v|u^p=v^q\rangle$ respectively. For $i\in\{1,2\}$, $x\in\{u,v\}$, and an integer $n$, define an element $\lambda(i,x,n)$ of $G$ as follows:

$$\lambda(1,x,n)=l_2^nx\bar{l}_2^n,\qquad \lambda(2,x,n)=\mu l_2^nx\bar{l}_2^n\bar{\mu}. \tag{4.7}$$

Then we see the $\Lambda$-invariant of the unknotting tunnel $\tau(i,x,n)$ is represented by $\lambda(i,x,n)$. Now the proof of Theorem 4.1 is reduced to the following Lemma.

(4.8) **Lemma.** (1) *Suppose* $\mathscr{P}_1\lambda(i,x,n)\mathscr{P}_1=\mathscr{P}_1\lambda(i,x,n')^\varepsilon\mathscr{P}_1$ *for some* $\varepsilon=\pm 1$, *then either* $n=n'$ *or* $\beta\equiv\pm 1 \bmod\alpha$.

(2) *Suppose* $\mathscr{P}_1\lambda(1,x,n)\mathscr{P}_1=\mathscr{P}_1\lambda(2,x,n')^\varepsilon\mathscr{P}_1$ *for some* $\varepsilon=\pm 1$, *then* $\beta\equiv\pm 1 \bmod\alpha$.

(3) $\mathscr{P}_1\lambda(i,u,n)\mathscr{P}_1\neq\mathscr{P}_1\lambda(i',v,n')^\varepsilon\mathscr{P}_1$.

*Proof.* (1) We prove this for the case where $i=1$ and $x=u$. Suppose

$$\mathscr{P}_1\lambda(1,u,n)\mathscr{P}_1=\mathscr{P}_1\lambda(1,u,n')^\varepsilon\mathscr{P}_1;$$

that is, there are elements $\sigma_1$ and $\sigma_2$ of $\mathscr{P}_1$ such that

$$\lambda(1,u,n)=\sigma_1\lambda(1,u,n')^\varepsilon\sigma_2 \quad \text{in} \quad G=\pi_1(E(K(\alpha,\beta;p,q)),a_1).$$

Then there is a continuous map $\Phi:[0,1]\times[0,1]\to E(K(\alpha,\beta;p,q))$ which satisfies the following conditions:

(i) $\Phi(\{0\}\times[0,1])=\Phi(\{1\}\times[0,1])=a_1$.

(ii) $\Phi|_{[0,1]\times\{t\}}$ represents $\lambda(1,u,n)$ or $\sigma_1\lambda(1,u,n')^\varepsilon\sigma_2$ according as $t=0$ or 1.

By noting (4.7), we may assume further that for $t=0$ and 1,

$$\Phi^{-1}(E(K_0))\cap[0,1]\times\{t\}=[\tfrac{1}{3},\tfrac{2}{3}]\times\{t\},\qquad \Phi(\{\tfrac{1}{3},\tfrac{2}{3}\}\times t)=a_2,$$

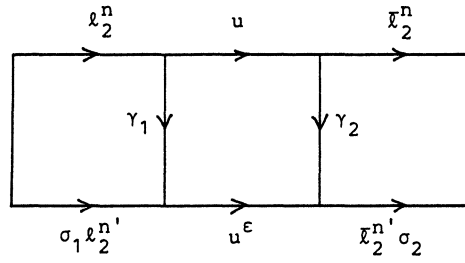



**Fig. 4.1**

and $\Phi|_{[\frac{1}{3},\frac{2}{3}]\times\{t\}}$ represents the element $u$ or $u^\varepsilon$ in $G(K_0)\cong\pi_1(E(K_0),a_2)$ according as $t=0$ or 1. Since $T$ is incompressible, and the elements $u$ and $u^\varepsilon$ are not peripheral in $E(K_0)$, we may assume $\Phi^{-1}(T_2)=\{\frac{1}{3},\frac{2}{3}\}\times[0,1]$. Let $\gamma_1$ and $\gamma_2$ be the elements of $\mathscr{P}_2=\pi_1(T_2,a_2)$ represented by $\Phi|_{\{\frac{1}{3}\}\times[0,1]}$ and $\Phi|_{\{\frac{2}{3}\}\times[0,1]}$ respectively (cf. Fig. 4.1).

Then we obtain the following identities.

$$\bar{\gamma}_1 u\gamma_2=u^\varepsilon \quad \text{in} \quad G(K_0). \tag{4.9}$$

$$l_2^n\gamma_1\bar{l}_2^{n'}\bar{\sigma}_1=l_2^n\gamma_2\bar{l}_2^{n'}\sigma_2=1 \quad \text{in} \quad G(L). \tag{4.10}$$

Suppose $\varepsilon=+1$. Then, by Lemma 4.5(1) and (4.9), we have $\gamma_1=\gamma_2=h^b=m_2^b$ for some integer $b$. So $\sigma_1=l_2^{n-n'}m_2^b$ by (4.10). Thus if $n\neq n'$, then $\sigma_1$ is a non-trivial element which belongs to $\mathscr{P}_1\cap\mathscr{P}_2$; so we have $\beta\equiv\pm1 \bmod\alpha$ by Lemma 4.4(2). Suppose $\varepsilon=-1$. Then, by Lemma 4.5(1) and (4.9), we have $\gamma_1=m_0h^{1-b}=l_2m_2^{1-b}$ and $\gamma_2=\bar{m}_0\bar{h}^b=\bar{l}_2\bar{m}_2^b$ for some integer $b$. So, (4.10) implies $\sigma_1=l_2^{n-n'+1}m_2^{1-b}$ and $\sigma_2=l_2^{-n+n'+1}m_2^b$. Thus $\sigma_1$ and $\sigma_2$ belong to $\mathscr{P}_1\cap\mathscr{P}_2$, and at least one of them is non-trivial. Hence $\beta\equiv\pm1 \bmod\alpha$ by Lemma 4.4(2). This completes the proof of Lemma 4.8(1) for the case where $i=1$ and $x=u$. The other cases can be settled similarly.

(2) Suppose $\mathscr{P}_1\lambda(1,u,n)\mathscr{P}_1=\mathscr{P}_1\lambda(2,u,n')^\varepsilon\mathscr{P}_1$. Then by an argument similar to the proof of (1), there are elements $\sigma_1,\sigma_2\in\mathscr{P}_1$ and $\gamma_1,\gamma_2\in\mathscr{P}_2$ which satisfy the following identities:

$$\bar{\gamma}_1 u\gamma_2=u^\varepsilon \quad \text{in} \quad G(K_0). \tag{4.11}$$

$$l_2^n\gamma_1\bar{l}_2^{n'}\bar{\mu}\bar{\sigma}_1=l_2^n\gamma_2\bar{l}_2^{n'}\bar{\mu}\sigma_2=1 \quad \text{in} \quad G(L). \tag{4.12}$$

Suppose $\varepsilon=+1$, then by Lemma 4.5(1), we have $\gamma_1=\gamma_2=m_2^b$ and $l_2^{n-n'}m_2^b\bar{\mu}\bar{\sigma}_1=1$. Hence we have $\beta\equiv\pm1 \bmod\alpha$ by Lemma 4.4(1). If $\varepsilon=-1$, then we also have the same conclusion by a similar argument. This completes the proof of Lemma 4.8(2) for the case where $x=u$. The case where $x=v$ can also be settled similarly.

(3) Suppose $\mathscr{P}_1\lambda(i,u,n)\mathscr{P}_1=\mathscr{P}_1\lambda(i',v,n')^\varepsilon\mathscr{P}_1$. Let $w$ and $w'$ be the elements of $G(L)$ such that $\lambda(i,u,n)=wu\bar{w}$ and $\lambda(i',v,n')=w'v\bar{w}'$ [see (4.7)]. Then there are elements $\sigma_1,\sigma_2\in\mathscr{P}_1$ and $\gamma_1,\gamma_2\in\mathscr{P}_2$ which satisfy the following identities:

$$\bar{\gamma}_1 u\gamma_2=v^\varepsilon \quad \text{in} \quad G(K_0). \tag{4.13}$$

$$w\gamma_1\bar{w}'\bar{\sigma}_1=w\gamma_2\bar{w}'\sigma_2=1 \quad \text{in} \quad G(L). \tag{4.14}$$

By Lemma 4.5(1) and (4.13), we see $q=p-1, \varepsilon=+1, \gamma_1=\bar{m}_2^b l_2$ and $\gamma_2=\bar{m}_2^b$ for some integer $b$. So, (4.14) implies $w\bar{m}_2^b l_2 \bar{w}' \bar{\sigma}_1 = w\bar{m}_2^b \bar{w}' \sigma_2 = 1$. Thus we have $l_2 \bar{w}' \bar{\sigma}_1 = \bar{w}' \sigma_2$, that is, $w' l_2 \bar{w}' = \sigma_2 \sigma_1$. However, this is impossible by Lemma 4.4(2) and (3). This completes the proof of Lemma 4.8.

To prove Theorem 4.2, we study the mapping class group $\mathcal{M}$ of $E(K(\alpha, \beta; p, q))$. Let $\mathcal{M}(L)$ and $\mathcal{M}(K_0)$ be the mapping class group of $E(L)$ and $E(K_0)$, and put

$$\Delta = \{([f_1], [f_2]) \in \mathcal{M}(L) \times \mathcal{M}(K_0) \mid f_1|_{T_2} \text{ is isotopic to } f_2|_{T_2}\}.$$

Let $\mathcal{D}$ denote the subgroup of $\mathcal{M}(L)$ generated by the Dehn twists along the essential torus $T_2$. Then we have

$$(4.15)\quad \mathcal{D} \cong \pi_1(T_2)/\langle Z(\pi_1(E(L))) + Z(\pi_1(E(K_0)))\rangle \cong \begin{cases} Z & \text{if } \beta \not\equiv \pm 1 \bmod \alpha, \\ \{1\} & \text{if } \beta \equiv \pm 1 \bmod \alpha. \end{cases}$$

Here $Z(\ )$ denotes the center of a group. Since $T_2$ gives the torus decomposition of $E(K(\alpha, \beta; p, q))$, we have the following exact sequence (cf. [BS, Sa 2]):

$$1 \to \mathcal{D} \to \mathcal{M} \to \Delta \to 1.$$

By using the calculation of the symmetry groups of 2-bridge links by [BS], we see

$$\Delta \cong \begin{cases} Z_2 & \beta^2 \not\equiv 1 \bmod \alpha, \\ Z_2 \oplus Z_2 & \beta^2 \equiv 1 \bmod \alpha. \end{cases}$$

We describe the generators of $\Delta$. We see $(S^3, K(\alpha, \beta; p, q))$ admits an orientation preserving involution $h$, such that $h(T_2)=T_2$, $h(l_i)=-l_i$ $(i=1,2)$, and $h|_{E(K_0)}$ is fibre-preserving and reverse the fibre-orientation. This gives the generator of $\Delta \cong Z_2$ for the case $\beta^2 \not\equiv 1 \bmod \alpha$. Note that $h|_{E(L)}$ preserves $\tau_1$ and $\tau_2$, and $h|_{E(K_0)}$ preserves each of the two singular fibres. If $\beta^2 \equiv 1 \bmod \alpha$, then there is an orientation preserving involution $f$ on $S^3$ such that $f(\tau_1)=\tau_2$, $f(K_2)=K_2$ and $f(K_1)=K_1$ or $-K_1$ according as $\beta^2 \equiv 1$ or $\alpha+1 \bmod 2\alpha$. Then $([f], [\mathrm{id}])$ generates the additional $Z_2$-factor of $\Delta$. Now the homeomorphism classification Theorem 4.2 follows from the above facts and the isotopy classification Theorem 4.1.

## 5. Detecting inequivalences of the unknotting tunnels for 2-bridge knots

In this section, we quickly recover the homeomorphism classification due to [BM, F, K 3] of the unknotting tunnels $\{\tau_i, \varrho_i, \varrho_i' (i=1,2)\}$ for the nontrivial 2-bridge knot $K=S(\alpha, \beta)$, and then give the isotopy classification of them.

(5.1) **Lemma.** *For the continued fraction expansion of $\alpha/\beta$ given in* 1.6, *we have the following:*

(1) $\beta \equiv \pm 1 \bmod \alpha$, *if and only if* $(a_1, a_2, \ldots, a_{2g-1}, a_{2g}) = \pm(1, -1, \ldots, 1, -1)$.
(2) $\beta \equiv \pm 2 \bmod \alpha$, *if and only if* $g=1$ *and* $|a_2|=1$.
(3) $\beta \equiv \pm 2^{-1} \bmod \alpha$, *if and only if* $g=1$ *and* $|a_1|=1$.
(4) $\beta^2 \equiv 1 \bmod \alpha$, *if and only if* $a_i = -a_{2g+1-i}$ $(1 \leqq i \leqq 2g)$.
(5) $\beta^2 \equiv -1 \bmod \alpha$, *if and only if* $a_i = a_{2g+1-i}$ $(1 \leqq i \leqq 2g)$.
(6) $\alpha=3$ *if and only if* $g=1$ *and* $(a_1, a_2) = \pm(1, -1)$.

*Proof.* See [HK, Lemma 2] and [Sa 1, p. 188].

(5.2) **Theorem.** *The following Tables A and B respectively give the isotopy and homeomorphism classification of the unknotting tunnels* $\{\tau_i, \varrho_i, \varrho_i'\ (i=1,2)\}$. *Two tunnels are isotopic (homeomorphic resp.) if and only if a condition in Lemma 5.1 whose number is contained in the corresponding box in Table A (B resp.) is satisfied.*

**Table A**

| $\tau_2$ | $\varrho_1$ | $\varrho_1'$ | $\varrho_2$ | $\varrho_2'$ | |
|---|---|---|---|---|---|
| 1 | 6 | 6 | 2 | 2 | $\tau_1$ |
| | 3 | 3 | 6 | 6 | $\tau_2$ |
| | | 1,3 | 1 | 1 | $\varrho_1$ |
| | | | 1 | 1 | $\varrho_1'$ |
| | | | | 1,2 | $\varrho_2$ |

**Table B**

| $\tau_2$ | $\varrho_1$ | $\varrho_2$ | |
|---|---|---|---|
| 4,5 | 6 | 2 | $\tau_1$ |
| | 3 | 6 | $\tau_2$ |
| | | 1 | $\varrho_1$ |

*Proof.* Since the "if" part is easily seen (cf. [K 3, Fig. 8], [Sa 1, Fig. 3.2]), we prove the "only if" part. Except the isotopy problem between $\varrho_i$ and $\varrho_i'$ $(i=1,2)$, the isotopy classification follows from Example 3.4 and Lemma 5.1. The homeomorphism classification follows from this isotopy classification and Schubert's result [Sb]. In fact, it follows from [Sb] that every homeomorphism of $(S^3, S(\alpha,\beta))$ is isotopic to a homeomorphism which maps $(B_1,t_1)$ to $(B_i,t_i)$ for $i=1$ or 2, and there is a homeomorphism mapping $(B_1,t_1)$ to $(B_2,t_2)$, if and only if $\beta^2 \equiv \pm 1 \bmod \alpha$ (cf. [BS]).

In the following, we prove that the tunnels $\varrho_1$ and $\varrho_1'$ ($\varrho_2$ and $\varrho_2'$ resp.) are isotopic, if and only if the condition (1) or (3) [(2) resp.] in Lemma 5.1 is satisfied.

To prove this, consider the Seifert surface $F \subset E(K)$ for $K$ as illustrated in Fig. 5.1. Then $l=\partial F \subset E(K)$ forms a longitude of $K$. Choose a meridian $m \subset \partial E(K)$ of $K$ so that $l \cap m$ is the point $p \in \partial E(K)$ as indicated in Fig. 5.1. Note that $F$ is obtained from $2g$ unknotted annuli $A_i$ in $S^3$ $(1 \leqq i \leqq 2g)$ with $(-1)^{i-1}a_i$ full twists through plumbing. Let $\alpha_i$ be an oriented loop forming a core of $A_i$, and let $\beta_1$ be an arc in $F$ joining $p$ and $\alpha_1 \cap \alpha_2$ as illustrated in Fig. 5.1. Consider the element of $\pi_1(F,p)$ represented by the oriented loop $\alpha_i$ together with an arc $\beta_i$ joining $\alpha_i$ and $p$, where $\beta_i$ is chosen so that

$$\beta_i \subset \beta_1 \cup \{\alpha_j \mid 2 \leqq j \leqq i-1\}$$

and $\beta_i$ does not go through the "twisted part" of $A_j$ $(2 \leqq j \leqq i-1)$. We denote this element by the same symbol $\alpha_i$. Then $\pi_1(F,p)$ is the free group $\langle \alpha_1, \ldots, \alpha_{2g} \rangle$ of rank $2g$. Let $E(F)$ be the manifold obtained from $E(K)$ by cutting along $F$, and let $q$ be the arc in $\partial E(F)$ obtained from $m$. Then $\pi_1(E(F),q)$ is the free group $\langle x_1, \ldots, x_{2g} \rangle$ of rank $2g$, where $x_i$ is the element of $\pi_1(E(F),q)$ as indicated in Fig. 5.1. Let $j_+$ and $j_-$ be the homomorphisms from $\pi_1(F,p)$ to $\pi_1(E(F),q)$ obtained by shifting each element of $\pi_1(F,p)$ to $+$ and $-$ sides respectively. Then we see

$$j_+(\alpha_{2i-1}) = x_{2i}^{a_{2i}-1}, \quad j_+(\alpha_{2i}) = \bar{x}_{2i-1}\bar{x}_{2i}^{a_{2i}}x_{2i+1}, \tag{5.3}$$

$$j_-(\alpha_{2i-1}) = x_{2i-2}x_{2i}^{a_{2i}-1}\bar{x}_{2i}, \quad j_-(\alpha_{2i}) = \bar{x}_{2i}^{a_{2i}}. \tag{5.4}$$

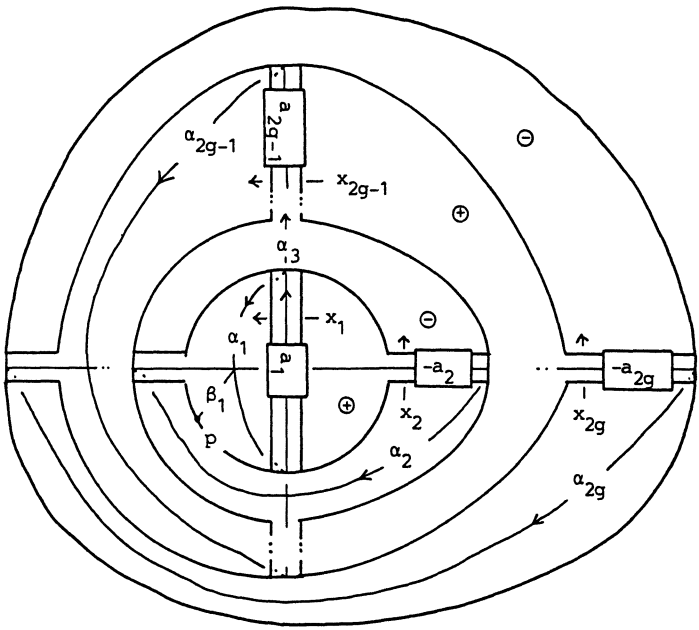



Fig. 5.1

Here $1 \leqq i \leqq 2g$ and $x_0 = x_{2g+1} = 1$. We denote the element $j_+(l) = j_-(l) \in \pi_1(E(F), q)$ by the same symbol $l$. Then it is given by

(5.5) $$l = W_0 \bar{W}_a W_e, \quad \text{where}$$

$$W_0 = \prod_{i=1}^{g} x_{2i-1}^{a_{2i-1}}, \qquad W_a = \prod_{i=1}^{g} x_{2i-1}^{a_{2i-1}} \bar{x}_{2i}^{a_{2i}}, \qquad W_e = \prod_{i=1}^{g} x_{2i}^{a_{2i}}.$$

Put $\lambda = x_1$ and $\lambda' = \bar{W}_e x_1 W_e$. Then we see the double coset invariants $\Lambda(\varrho_1)$ and $\Lambda(\varrho_1')$ are represented by $\lambda$ and $\lambda'$ respectively (see Fig. 5.2).

(5.6) **Lemma.** *$\lambda^\varepsilon$ and $\lambda'$ represent the same double coset of $\mathscr{P}(K)$ in $G(K)$ for some $\varepsilon = \pm 1$, if and only if one of the following conditions holds:*

(1) *$\lambda^\varepsilon$ and $\lambda'$ represent the same double coset of $\langle l \rangle$ in $\pi_1(E(F), q)$.*

(2) *There are elements $\gamma_1, \ldots, \gamma_n$ of $\pi_1(F, p)$, such that*

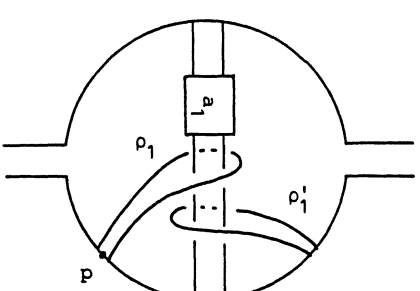



Fig. 5.2

(i) $\lambda^{\varepsilon}=j_{-}(\gamma_1)$ *and* $j_{+}(\gamma_i)=j_{-}(\gamma_{i+1})$ $(1\leqq i\leqq n-1)$ *in* $\pi_1(E(F),q)$, *and*
(ii) $j_{+}(\gamma_n)$ *and* $\lambda'$ *represent the same double coset of* $\langle l\rangle$ *in* $\pi_1(E(F),q)$.
(3) *There are elements* $\gamma_1,\ldots,\gamma_n$ *of* $\pi_1(F,p)$ *such that*
(i) $\lambda^{\varepsilon}=j_{+}(\gamma_1)$ *and* $j_{-}(\gamma_i)=j_{+}(\gamma_{i+1})$ $(1\leqq i\leqq n-1)$ *in* $\pi_1(E(F),q)$, *and*
(ii) $j_{-}(\gamma_n)$ *and* $\lambda'$ *represent the same double coset of* $\langle l\rangle$ *in* $\pi_1(E(F),q)$.

*Proof.* Suppose $\lambda^{\varepsilon}$ and $\lambda'$ represent the same double coset of $\mathscr{P}(K)$ of $G(K)$, that is,

$$(m^a l^b)\lambda^{\varepsilon}(m^c l^d)=\lambda'$$

for some integers $a, b, c$, and $d$. Considering this identity in $H_1(E(K))$, we see $c=-a$. So, we have

$$m^a\lambda^{\varepsilon}\bar{m}^a=\bar{l}^b\lambda'\bar{l}^d \tag{5.7}$$

Let $\varphi_0$ ($\varphi_1$ resp.) be a continuous map from $([0,1],\partial[0,1])$ to $(E(K),p)$ representing the left (right resp.) hand side of (5.7). Since $\bar{l}^b\lambda'\bar{l}^d$ is contained in $\pi_1(E(F),q)$, we may assume there is a surface $F'$ in $E(F)$ which is parallel to $F$, such that $\varphi_1^{-1}(F')=\emptyset$. Put $n=|a|$ and $p'=m\cap F'$. Then we may assume (i) $\varphi_0^{-1}(F')=\varphi_0^{-1}(p')=\{t_i|1\leqq i\leqq 2n\}$, where $0<t_1<t_2<\ldots<t_{2n}<1$, and (ii) if $n\geqq 1$, then $\varphi_0([0,t_1])$, $\varphi_0([t_i,t_{i+1}])$ $(i\neq n)$ and $\varphi_0[t_{2n},1]$ are contained in the loop $m$. By (5.7), there is a continuous map $\Phi:[0,1]\times[0,1]\to E(K)$, such that $\Phi_{[0,1]\times\{i\}}=\varphi_i$ $(i=0,1)$, and $\Phi(\{0,1\}\times[0,1])=\{p\}$. By using the fact that $F$ is incompressible, we may assume $\Phi^{-1}(F')$ consists of mutually disjoint proper arcs $\gamma_i$ $(1\leqq i\leqq n)$, such that $\partial\gamma_i=\{t_{n+1-i},t_{n+i}\}$ (see Fig. 5.3). If $n=0$, then the condition (1) holds. If $n\neq 0$, then let $\gamma_i$ be the element of $\pi_1(F',p')\cong\pi_1(F,p)$ represented by $\Phi|_{\gamma_i}$. Then we see the condition (2) or (3) holds according as $a$ is positive or negative.

(5.8) **Lemma.** *The condition* (1) *in Lemma 5.6 holds, if and only if the condition* (3) *in Lemma 5.1 holds.*

*Proof.* Recall that $\pi_1(E(F),q)$ is the free group with free basis $\{x_i|1\leqq i\leqq 2g\}$. By using (5.5), we see the length of the word $l^r\lambda'l^s$ is 1, if and only if one of the following holds:

(i) $g=1$, $a_1=1$, and $(r,s)=(1,0)$.
(ii) $g=1$, $a_1=-1$, and $(r,s)=(0,-1)$.

In these cases, $l^r\lambda'l^s$ is equal to $x_1=\lambda$. This completes the proof of Lemma 5.8.

In the following we prove that the condition (2) or (3) in Lemma 5.6 holds, if and only if the condition (1) in Lemma 5.1 holds. To do this we need the following lemma.

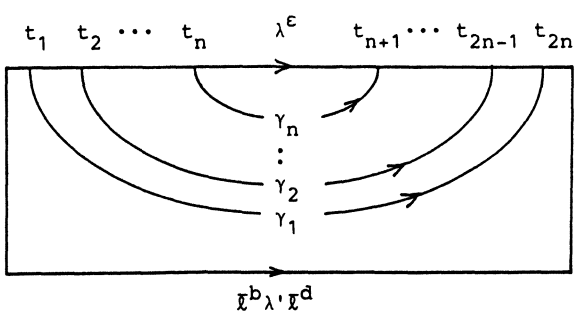



**Fig. 5.3**

(5.9) **Lemma.** *Let $n$ be an integer such that $0<n\leqq g$. Then:*

(1) $x_{2n-1}\in \mathrm{Im} j_-$, *if and only if* $|a_i|=1$ $(2n-2\leqq i\leqq 2n)$. *Further, if this condition is satisfied, then we have*

$$x_{2k}=j_-(\bar{\alpha}_{2k}^{a_{2k}}) \quad (k=n-1,n),$$

$$x_{2n-1}=j_-((\alpha_{2n-2}^{a_{2n-2}}\alpha_{2n-1}\bar{\alpha}_{2n}^{a_{2n}})^{a_{2n-1}}).$$

(2) $x_{2n}\in \mathrm{Im} j_+$, *if and only if* $|a_i|=1$ $(2n-1\leqq i\leqq \min\{2n+1,2g\})$. *Further, if this condition is satisfied, then we have*

$$x_{2k-1}=j_+(\alpha_{2k-1}^{a_{2k-1}}) \quad (k=n,n+1),$$

$$x_{2n}=j_+((\alpha_{2n-1}^{a_{2n-1}}\alpha_{2n}\bar{\alpha}_{2n+1}^{a_{2n+1}})^{-a_{2n}}).$$

*Here $\alpha_0$ and $\alpha_{2g+1}$ denote the trivial element* 1.

(3) *If* $|a_i|=1$ $(1\leqq i\leqq 2g)$, *then* $\mathrm{Im} j_+=\mathrm{Im} j_-=\pi_1(E(F),q)$.

*Proof.* (1) and (2) can be proved by studying the abelianizations of the homomorphisms $j_+$ and $j_-$. (3) follows from (1) and (2).

(5.10) **Lemma.** *Let $n$ be a positive integer. Then:*

(1) *There are elements $\gamma_1,\ldots,\gamma_n$ of $\pi_1(F,p)$, such that*

$$x_1=j_-(\gamma_1) \quad \text{and} \quad j_+(\gamma_i)=j_-(\gamma_{i+1})\ (1\leqq i\leqq n-1) \quad \text{in} \quad \pi_1(E(F),q),$$

*if and only if* $|a_i|=1$ $(1\leqq i\leqq \min\{2n,2g\})$. *Moreover, if this condition is satisfied and $n\leqq g$, then $\gamma_n$ is contained in the subgroup $\langle\alpha_1,\ldots,\alpha_{2n}\rangle$, and the letter $\alpha_{2n}$ appears in the reduced word representing $\gamma_n$ precisely once with weight $\pm 1$.*

(2) *There are elements $\gamma_1,\ldots,\gamma_n$ of $\pi_1(F,p)$, such that*

$$x_1=j_+(\gamma_1) \quad \text{and} \quad j_-(\gamma_i)=j_+(\gamma_{i+1})\ (1\leqq i\leqq n-1) \quad \text{in} \quad \pi_1(E(F),q),$$

*if and only if* $|a_i|=1$ $(1\leqq i\leqq \min\{2n-1,2g\})$. *Moreover, if this condition is satisfied and $n\leqq g$, then $\gamma_n$ is contained in the subgroup $\langle\alpha_1,\ldots,\alpha_{2n-1}\rangle$, and the letter $\alpha_{2n-1}$ appears in the reduced word representing $\gamma_n$ precisely once with weight $\pm 1$.*

*Proof.* (1) Since the "if" part is clear, we prove the "only if" part by induction on $n$. [The first step of the induction can be proved by an argument similar to the following.] Suppose there are elements $\gamma_1,\ldots,\gamma_n$ of $\pi_1(F,p)$, such that $x_1=j_-(\gamma_1)$ and $j_+(\gamma_i)=j_-(\gamma_{i+1})$ $(1\leqq i\leqq n-1)$ in $\pi_1(E(F),q)$. Then by the inductive hypothesis, we have $|a_i|=1$ $(1\leqq i\leqq 2(n-1))$ and $\gamma_{n-1}$ is expressed by a word of $\{\alpha_1,\ldots,\alpha_{2(n-1)}\}$ in which $\alpha_{2(n-1)}$ appears precisely once with weight $\pm 1$. Thus, by (5.3), we see $j_+(\gamma_{n-1})$ is expressed by a word of $\{x_1,\ldots,x_{2n-1}\}$ in which $x_{2n-1}$ appears precisely once with weight $\pm 1$. On the other hand, since $|a_i|=1$ $(1\leqq i\leqq 2(n-1))$, the subgroup $\langle x_1,\ldots,x_{2(n-1)}\rangle$ is contained in $\mathrm{Im} j_-$ by Lemma 5.9(1). Hence $j_+(\gamma_n)\in \mathrm{Im} j_-$, if and only if $x_{2n-1}\in \mathrm{Im} j_-$. This occurs if and only if $|a_i|=1$ $(2n-2\leqq i\leqq 2n)$ by Lemma 5.9(1). Thus we have $|a_i|=1$ $(1\leqq i\leqq 2n)$. Noting the form of the word representing $j_+(\gamma_{n-1})$ and identities in Lemma 5.9(1), we see $\gamma_n$ is represented by a word of $\{x_1,\ldots,x_{2n}\}$ in which $x_{2n}$ appears only once with weight $\pm 1$. This completes the proof of (1). (2) can be proved similarly by using Lemma 5.9(2).

(5.11) **Corollary.** *If the condition* (2) *or* (3) *in Lemma 5.6 is satisfied, then* $|a_i|=1$ $(1\leqq i\leqq 2g)$.

*Proof.* Suppose the condition (2) in Lemma 5.6 is satisfied. If $n \leqq g-1$, then by (5.3) and Lemma 5.10(1), $j_+(\gamma_n)$ is expressed by a word of $\{x_1, \ldots, x_{2n+1}\}$, where $x_{2n+1}$ appears precisely once with weight $\pm 1$. Thus $\pm[\lambda'] = \pm[x_1] \mp [j_+(\gamma_n)]$ in $H_1(E(F))$. Since $l$ is null-homologous, this implies that $\lambda'$ and $j_+(\gamma_n)$ determine different double cosets of $\langle l \rangle$ in $\pi_1(E(F))$. Hence $n \geqq g$, and we obtain the desired result by Lemma 5.10(1). Similarly, by using Lemma 5.10(2), we can prove the same result in case the condition (3) in Lemma 5.6 is satisfied.

In the following, we assume $|a_i| = 1$ $(1 \leqq i \leqq 2g)$. Then $K$ is a fibred knot with fibre $F$, whose monodromy $\varphi$ is given by $\varphi = D_1^{a_1} D_2^{-a_2} \ldots D_{2g-1}^{a_{2g-1}} D_{2g}^{-a_{2g}}$, where $D_i$ denotes the Dehn twist homeomorphism of $F$ along the simple loop $\alpha_i$ in Fig. 5.1 (cf. [Ga]). Let $\hat{F}$ be the closed surface of genus $g$ obtained by closing $F$, and let $\hat{\varphi}$ be a homeomorphism of $\hat{F}$ which is an extension of $\varphi$. Since $\pi_1(\hat{F}) \cong \pi_1(F)/\langle\langle l \rangle\rangle$, we obtain the following by using [E, Theorem 2.1].

(5.12) **Lemma.** *Suppose the condition* (2) *or* (3) *in Lemma 5.6 is satisfied. Then there is a non-zero integer $n$ such that the simple loop $\hat{\varphi}^n(\alpha_1)$ is isotopic to $\alpha_1$ in $\hat{F}$.*

The simple loops $\alpha_i$ $(1 \leqq i \leqq 2g)$ are situated in $\hat{F}$ as illustrated in Fig. 5.4, and they are invariant by the involution $\iota$ as illustrated in Fig. 5.4. Hence $\hat{\varphi}$ commutes with $\iota$, and it induces a homeomorphism $\psi$ of $(\hat{F}, Fix(\iota))/\iota \cong (S^2, P)$, where $P$ consists of $2g+2$ points $\{p_i \mid 1 \leqq i \leqq 2g+2\}$. Let $M(0, 2g+2)$ be the mapping class group of $(S^2, P)$, and let $\omega_i$ be a twist homeomorphism with support a regular neighbourhood of the arc $\beta_i \subset S^2$ as illustrated in Fig. 5.4, which is the image of $\alpha_i$ (cf. [Bi, p. 164]). Then as an element of $M(0, 2g+2)$, we have

$$\psi = \omega_1^{a_1} \omega_2^{-a_2} \ldots \omega_{2g-1}^{a_{2g-1}} \omega_{2g}^{-a_{2g}}.$$

Note that the permutation on $P$ induced by $\psi$ is $(2g+1, 2g, \ldots, 2, 1)(2g+2)$.

(5.13) **Lemma.** *$\psi$ has finite order in $M(0, 2g+2)$, if and only if the condition* (1) *in Lemma 5.1 is satisfied.*

*Proof.* Suppose $\psi$ has finite order. Then its order is a multiple of $2g+1$ by the remark stated in the above. On the other hand, the order of a torsion element of

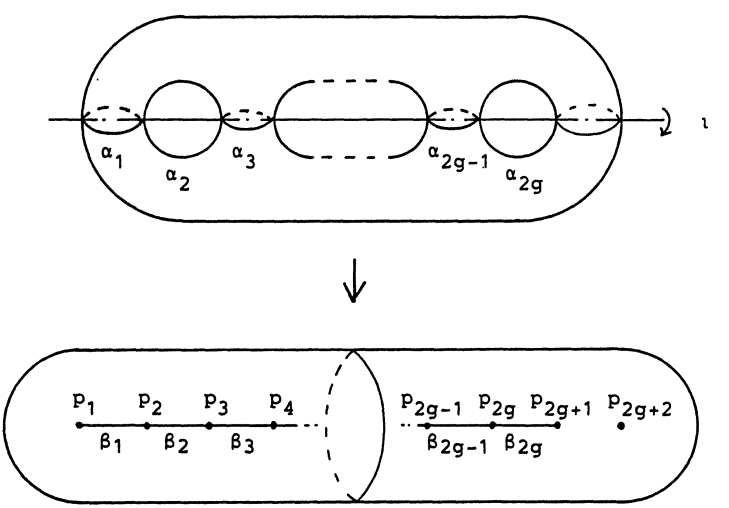



**Fig. 5.4**

$M(0, 2g+2)$ is a divisor of $2g+2$, $2g+1$, or $2g$ by [Mu, Theorem D and Remark in p. 77]. Hence $\psi$ has order $2g+1$, and by [Mu, Theorem D], we see $\psi$ is conjugate to $\psi_0^k$, where $\psi_0=\omega_1\omega_2\ldots\omega_{2g}$ and $k$ is an integer relatively prime to $2g+1$. Note that $\psi_0$ is equal to the "$\psi$" for the case where $(a_1, a_2, \ldots, a_{2g-1}, a_{2g})=(1, -1, \ldots, 1, -1)$. Thus we have only to show that $\psi$ is not conjugate to $\psi_0^k$ if

$$(a_1, a_2, \ldots, a_{2g-1}, a_{2g}) \neq \pm(1, -1, \ldots, 1, -1).$$

Note that $\hat{\phi}$ is a lift of $\psi$, and $|\mathrm{trace}\,\hat{\phi}_*|$ is an invariant of the conjugacy class of $\psi$, where $\hat{\phi}_*$ is the automorphism of $H_1(\hat{F})$ induced by $\hat{\phi}$. Then the desired result is deduced from the following fact:

(5.14) **Sublemma.** (1) $\mathrm{trace}\,\hat{\phi}_*=2g+\sum_{i=1}^{2g-1} a_i a_{i+1}$. *In particular,* $|\mathrm{trace}\,\hat{\phi}_*|=1$, *if and only if* $(a_1, a_2, \ldots, a_{2g-1}, a_{2g})=\pm(1, -1, \ldots, 1, -1)$.

(2) $|\mathrm{trace}\,\hat{\phi}_{0*}^k|=1$ *where* $k$ *is an integer relatively prime to* $2g+1$, *and* $\hat{\phi}_0$ *is the "*$\hat{\phi}$*" for the case where* $(a_1, a_2, \ldots, a_{2g-1}, a_{2g})=\pm(1, -1, \ldots, 1, -1)$.

*Proof.* (1) can be obtained by direct calculation.

(2) Let $\mathscr{R}(k)$ be the set of all roots of the characteristic polynomial of $\hat{\phi}_{0*}^k$. Since the characteristic polynomial of $\hat{\phi}_{0*}$ is the Alexander polynomial of the torus knot $T(2g+1, 2)$, we have

$$\mathscr{R}(1)=\{\xi \mid \xi^{2(2g+1)}=1, \xi^2 \neq 1, \xi^{2g+1} \neq 1\}.$$

Noting that $k$ is relatively prime to $2g+1$ and $\mathscr{R}(k)=\{\xi^k \mid \xi \in \mathscr{R}(1)\}$, we can see $\mathrm{trace}\,\hat{\phi}_{0*}^k=(-1)^{k+1}\mathrm{trace}\,\hat{\phi}_{0*}$.

We can now prove the following lemma, which completes the proof of Theorem 5.2.

(5.15) **Lemma.** *If the condition* (2) *or* (3) *in Lemma 5.6 is satisfied, then*

$$(a_1, a_2, \ldots, a_{2g-1}, a_{2g})=\pm(1, -1, \ldots, 1, -1).$$

*Proof.* Let $\bar{\psi}$ be a homeomorphism of the sphere with $2g+2$ holes $\mathrm{cl}(S^2-N(P))$ obtained from $\psi$, and let $\beta_1'$ be the essential loop in $\mathrm{cl}(S^2-N(P))$ which forms the boundary of a regular neighbourhood of the arc $\beta_1$ in Fig. 5.4. Suppose the condition (2) or (3) in Lemma 5.7 is satisfied. Then there is a non-zero integer $n$, such that $\hat{\phi}^n(\alpha_1)$ is isotopic to $\alpha_1$ by Lemma 5.12. By [BH, Proof of Theorem 5], we can make the above isotopy $\iota$-equivariant. Hence the arc $\psi^n(\beta_1)$ is isotopic to $\beta_1$ rel. $\partial\beta_1$, and therefore $\bar{\psi}^n(\beta_1')$ is isotopic to $\beta_1'$. Hence $\bar{\psi}$ is periodic or reducible in the sense of [Th] (cf. [Hn]). Suppose

$$(a_1, a_2, \ldots, a_{2g-1}, a_{2g}) \neq \pm(1, -1, \ldots, 1, -1).$$

Then $\bar{\psi}$ is not isotopic to a periodic map by Lemma 5.13. Thus $\bar{\psi}$ must be reducible. On the other hand, we see every essential loop in $\mathrm{cl}(S^2-N(P))$ intersects with $\bar{\psi}^i(\beta_1')$ for some $i$ $(0 \leqq i \leqq 2g)$. [In fact, the union of the arcs $\psi^i(\beta_1)$ $(0 \leqq i \leqq 2g)$ join the points $p_i$ $(1 \leqq i \leqq 2g+1)$. Hence if a simple loop in $\mathrm{cl}(S^2-N(P))$ does not intersect $\bigcup_{i=0}^{2g} \bar{\psi}^i(\beta_1')$, then it is either contractible or isotopic to the boundary of a regular neighbourhood of $p_{2g+2}$.] So $\bar{\psi}$ cannot be reducible by [Hn, Lemma 2.2]. This is a contradiction. Hence we obtain the desired result.

(5.16) *Remark.* Uchida [U] gave an alternative proof to the discrimination of the tunnels $\varrho_i$ and $\varrho_i'$ $(i=1, 2)$ using dihedral branched coverings.

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